Simulation of a Modular BBM92 Receiver Subsystem

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1 Introduction

1.1 Context and Problem Statement

Quantum Key Distribution (QKD) represents a paradigm shift in secure communications, offering information-theoretic security guaranteed by the laws of physics rather than computational complexity. While optical fibre is the standard medium for metropolitan-scale quantum networks, its reach is fundamentally limited by exponential attenuation, typically restricting transmission to a few hundred kilometres without the use of quantum repeaters [1]. To achieve global-scale connectivity, satellite-based QKD has emerged as the leading solution, leveraging the vacuum of space to minimize optical loss over intercontinental distances [2].

However, satellite links introduce unique challenges compared to static fibre connections. The optical channel is dynamic, subject to rapid fluctuations in loss due to geometric diffraction, pointing errors, and atmospheric turbulence [3]. In this high-loss regime, the performance of the system is critically dependent on the receiver subsystem, particularly the detection hardware. As identified in recent experimental missions, the choice between standard Silicon Avalanche Photodiodes (Si-APDs) and emerging Superconducting Nanowire Single-Photon Detectors (SNSPDs) determines the feasibility of generating a secure key [4, 5].

1.2 Project Objective

The primary objective of this project is to develop a modular software simulation of a BBM92 Receiver Subsystem. Unlike comprehensive link-budget models that estimate performance averages, this project focuses on the "ground segment" logic: the discrete detection of single photons, the filtering of background noise via coincidence matching, and the extraction of a secure key from raw timestamps.

The simulation is designed to be hardware-agnostic, allowing for a comparative analysis between:

- **Current Technology:** Si-APDs as used in the Micius satellite mission [5].
- **Future Technology:** SNSPDs, which offer lower noise floors and higher efficiencies suitable for extreme-loss scenarios [6].

1.3 Report Structure

The remainder of this report is organized as follows: Section 2 presents a literature review of satellite and fibre QKD, establishing the theoretical bounds and experimental benchmarks used to calibrate the simulation. Section 3 details the modular architecture of the receiver subsystem, including the specific logic used for coincidence matching and error estimation. Section 4 presents the simulation results, analyzing the secure key rate and Quantum Bit Error Rate (QBER) under varying loss conditions. Finally, Section 5 summarizes the findings and outlines the roadmap for future extensions of the modular framework.

2 Literature Review

This review examines the current state of satellite and fibre-based QKD to identify the critical parameters required for an accurate receiver simulation. Specifically, it analyses the operational constraints of the BBM92 protocol, the impact of channel loss on secure key rates, and the performance differences between current silicon avalanche photodiodes and emerging superconducting detectors. These findings directly inform the design of the modular receiver subsystem described in 3.

2.1 Satellite Quantum Key Distribution

A central reason satellites are useful for long-distance QKD is that most of the optical path between a satellite and a ground station lies in vacuum. In vacuum, photons do not get absorbed the way they do in a physical medium, so the main source of loss is simple geometric beam spreading. The transmitted beam gradually widens, and only a fraction of it enters the receiving telescope. A common way to

describe this is through a diffraction-limited expression such as

$$T_{\text{sat}} \simeq \left(\frac{D_r}{2w(L)}\right)^2 e^{-\tau_{\text{atm}}}, \tag{1}$$

where D_r is the receiver aperture size, $w(L)$ is the beam radius after distance L, and $e^{-\tau_{\text{atm}}}$ represents attenuation in the atmosphere. The atmosphere only affects the last segment of the path, so the overall loss increases slowly with distance compared to ground-based links [7, 8]. This is what allows satellites in low Earth orbit to send quantum signals over thousands of kilometres without the loss becoming overwhelming.

Because the loss is mainly geometric, the practical limitations are not distance itself but the things that distort or block the beam: turbulence, cloud cover, pointing errors, and background light. These are what set the real operational boundaries of satellite QKD [3, 2].

2.1.1 Satellite Link Configurations

There are three main configurations used or proposed for satellite QKD:

- **Downlinks:** The satellite sends photons to the ground. This is usually the most favourable option because the photons only encounter turbulence near the ground, and the source is placed in the more stable vacuum environment.
- **Uplinks:** The ground station sends photons to the satellite. This is technically harder because the photons pass through the turbulent atmosphere immediately after leaving the transmitter, so the beam spreads more strongly.
- **Inter-satellite links:** Photons are exchanged between satellites in space. Since the entire path is in vacuum, these links can in principle have very low loss, but require extremely precise pointing and synchronisation.

2.1.2 Protocols Used in Satellite QKD

The three protocols I studied all have slightly different properties, and each one fits the satellite setting differently.

BB84

BB84 is the most commonly implemented protocol in satellite experiments. The satellite (or sometimes the ground station) prepares single photons in one of four polarisation states and sends them to the receiver. BB84 works well in satellite scenarios because:

- it tolerates high losses as long as the quantum bit error rate is not too large,
- it does not require an entangled source, which simplifies the payload,
- decoy-state versions make it secure even with practical photon sources.

Most long-distance demonstrations to date used BB84, including the major LEO experiments [9].

E91

E91 is conceptually different: instead of the satellite preparing states to send, it creates entangled photon pairs and sends one photon to each of two ground stations. Because the security is guaranteed by the correlations between the measurement outcomes, the satellite does not need to be trusted. This is one of the biggest motivations for using entanglement in space [10, 11]. The downside is that entangled sources are usually less bright, and the loss occurs in *two* links instead of one. This makes the detected count rates very low unless detectors and optics are extremely efficient. Even so, successful long-distance demonstrations have shown that this is a realistic path toward eliminating the trusted-node assumption [11].

BBM91 BBM92 is the entanglement-based sibling of BB84. Instead of preparing individual photon states, the satellite produces entangled photon pairs and sends one photon to each user (or one to the ground and keeps one locally, depending on the setup). The users measure in randomly chosen bases, just like in BB84.

In practice, BBM92 tends to be easier to implement than E91 because it does not require Bell-test measurements — the users simply measure correlations in the BB84 bases. This simplifies things experimentally but also means the satellite does need to be trusted. However, it still suffers the entanglement-distribution problem: both photons need to survive their respective channels.

2.1.3 Operational Constraints

Satellite QKD faces several practical limitations that arise from the physics of free-space links and how satellites move.

- **Night-time operation:** Sunlight produces enormous background noise in the detectors. At night, this drops dramatically, making the quantum signals easier to distinguish. Daytime QKD is not impossible, but it requires very narrow spectral filters, tight spatial filtering, and short detection time windows, which reduce the rate.
- **Limited visibility windows:** A low Earth orbit satellite is only above a given ground station for a few minutes. This means key generation is intermittent, and keys must be stored for later use. Increasing the number of satellites or using higher orbits can help, but this increases launch costs.
- **Weather dependence:** Clouds physically block the photons. Turbulence can distort the wavefront, coupling less light into the receiving telescope. This is why many ground stations are placed in high-altitude, dry environments.
- **Pointing and tracking:** The satellite and ground station need to keep their telescopes aligned within a few microradians. Any misalignment spreads the beam out and increases loss. Advanced beacon lasers and fast steering mirrors are used to correct this in real time.

2.1.4 Security Considerations and the Trusted Node Issue

Most current satellite QKD systems operate as "trusted nodes." This means the satellite either holds part of the key or performs the measurements, so the users must assume that the satellite has not been compromised. This might be acceptable for communication within a single nation but is problematic for international use.

The reason trusted nodes appear in many designs is primarily technical:

- entanglement-based links require higher efficiency and brightness than is currently available in space,
- synchronising two separate ground downlinks with very weak entangled signals is difficult,
- detectors in orbit face radiation, which increases dark counts and affects key rates.

However, trusted nodes do *not* have to be permanent. They can be avoided by:

- using E91-like protocols with entanglement distribution,
- improving entangled photon source brightness,
- deploying radiation-hardened, high-efficiency detectors,
- using satellite constellations to increase the number of useful passes and average the loss.

Recent experiments have already demonstrated trustless entanglement distribution between distant ground stations [11], suggesting that fully trustless satellite networks may be achievable in the future.

2.1.5 Future Directions

A few technological developments are expected to play an important role in making satellite QKD more practical and less dependent on trusted nodes. Here I briefly explain what each improvement actually does and why it matters.

- **Adaptive optics (AO):** Even though most of the satellite–ground path is in vacuum, the last few kilometres go through the turbulent atmosphere. Turbulence distorts the wavefront of the incoming beam, which reduces how much light couples into the receiving telescope. Adaptive optics uses a deformable mirror and a reference beacon to measure and correct these distortions in real time.

AO can significantly increase the signal-to-noise ratio, especially for uplinks (which suffer the most from turbulence), and could make daytime QKD more realistic. The challenge is building AO systems that are fast and precise enough but still compact, robust, and power-efficient for space missions.

- **Brighter entangled photon sources:** Entanglement-based QKD currently struggles with low count rates because entangled photon sources are not very bright and because both photons must survive large channel losses. Space-qualified sources based on periodically poled nonlinear crystals or integrated photonics can increase the pair production rate by orders of magnitude. Brighter sources directly support trustless (E91-style) QKD and make multi-user or multi-ground-station links more feasible. The open challenge is producing high-brightness, high-stability sources that survive radiation, temperature cycling, and launch vibration.
- **Superconducting nanowire single-photon detectors (SNSPDs):** SNSPDs offer extremely high efficiency (above 90%), very low dark counts, and excellent timing resolution. These properties make them ideal for high-loss links such as satellite QKD. Using SNSPDs on satellites would reduce noise, increase secure key rates, and make entanglement distribution over long baselines more feasible. The challenge is that SNSPDs require cryogenic cooling, so space-compatible cryocoolers must be efficient and long-lived.
- **Satellite constellations:** One LEO satellite only gives a few minutes of contact with each ground station per orbit. A constellation—several satellites working together—can provide more frequent passes, better coverage, and redundancy against weather. Constellations also enable multi-hop architectures (satellite-to-satellite-to-ground), improving global coverage without requiring a direct line-of-sight between endpoints. Open challenges include routing, synchronisation, and precision pointing across many moving spacecraft.
- **Inter-satellite quantum links:** Linking satellites with quantum channels in space avoids atmospheric loss entirely. These links could form the backbone of a space-based quantum repeater network. If satellites can distribute entanglement between themselves, they can perform entanglement swapping and extend correlations over global distances. The technological hurdles involve extremely tight pointing requirements, long-distance synchronisation, and stable high-rate entangled sources in orbit.

These developments aim to make satellite QKD faster, more robust, less dependent on weather, and ultimately capable of supporting trustless global quantum communication networks.

2.1.6 Active Research Groups and Countries

Several countries and institutions are heavily involved in satellite QKD research:

- **China:** Led by the University of Science and Technology of China (USTC) and the Chinese Academy of Sciences, responsible for the Micius satellite [9, 11].
- **Europe:** Research at the University of Vienna and the Austrian Academy of Sciences (IQOQI) [12], the University of Geneva, and ESA under projects such as SAGA and the Quantum Flagship [13].
- **Canada:** The University of Waterloo and the Institute for Quantum Computing (IQC) collaborate with the Canadian Space Agency on the QEYSSat mission [14].
- **Singapore:** The Centre for Quantum Technologies (CQT) works on small-satellite demonstrations and international collaborations [15, 16].

2.2 Fibre Quantum Key Distribution

Optical fibre is the most mature and widely deployed physical channel for QKD. The main physical limitation is absorption in the fibre, which is described by the exponential attenuation law

$$T(L) = 10^{-\alpha L/10}, \tag{2}$$

where L is the length in kilometres and α is the attenuation coefficient in dB/km (around 0.2 dB/km at 1550 nm [1]). Because the loss increases exponentially, the detected photon rate drops quickly with

distance. Eventually, detector noise starts to dominate, causing the quantum bit error rate (QBER) to increase and the secret key rate to approach zero.

Unlike free-space links, fibre channels are extremely stable: no turbulence, no beam wandering, and no pointing issues. But the price for this stability is high attenuation, which fundamentally restricts how far QKD can go without quantum repeaters.

2.2.1 Characteristics of Fibre Channels

Several physical effects determine how QKD protocols behave in optical fibre:

- **Loss:** This is the dominant limitation. Once the channel loss exceeds roughly 40–50 dB, most QKD protocols struggle to produce secure key rates with today's detectors.
- **Polarisation changes:** Standard fibre does not preserve polarisation. Small mechanical stresses or temperature variations change the polarisation state unpredictably. This makes polarisation-based QKD difficult over long distances unless active compensation is continuously applied.
- **Chromatic dispersion:** Dispersion broadens pulses as they travel. For high-speed systems, this limits how well detectors can distinguish between the time slots used by the protocol. Dispersion compensation is usually required for distances beyond tens of kilometres.
- **Raman noise (important in real networks):** Raman scattering occurs when intense classical data channels running through the same fibre generate broadband noise photons by interacting with the glass molecules [17]. These noise photons:
  - overlap with the QKD wavelength band,
  - arrive randomly in time,
  - increase the QBER,
  - can entirely overwhelm the quantum signal.

  Raman noise is one of the biggest obstacles to integrating QKD with classical telecom traffic. To mitigate it, researchers use spectral filtering, lower classical-power levels, and wavelength separation between classical and quantum channels.

These effects define how fibre QKD must be engineered, and they motivate the use of certain encodings, most importantly time-bin encoding.

2.2.2 Time-bin Encoding

Because fibre does not preserve polarisation, many fibre-based QKD systems use **time-bin encoding**. The idea is simple:

- A photon is placed into a superposition of two well-defined time slots: an "early bin" and a "late bin".
- Logical states are represented by the presence of the photon in one bin or the other, or in a coherent superposition of both.
- At the receiver, an interferometer recombines these two paths, allowing measurement of the phase between the bins.

Time-bin encoding is very robust in fibre because:

- temporal modes are preserved much more reliably than polarisation,
- the interferometers used for creation and detection can be actively stabilised,
- it works naturally for both prepare-and-measure (BB84) and entanglement-based protocols (BBM92, E91).

Most long-distance fibre demonstrations of BB84 and almost all fibre demonstrations of E91 and BBM92 use time-bin entanglement [18].

2.2.3 Protocols Used in Fibre QKD

Here I focus on the three protocols relevant to my project: BB84, BBM92, and E91. They each have different trust assumptions and different sensitivities to fibre imperfections.

BB84

BB84 is the simplest protocol and the most commonly deployed. In fibre implementations:

- weak laser pulses are used as approximate single-photon sources,
- time-bin encoding is often used instead of polarisation,
- decoy states are added to detect photon-number-splitting attacks.

Trust assumptions: BB84 is *device dependent*. Alice must trust that her state preparation device emits the states she thinks it does. Bob must trust the behaviour of his detectors (e.g., that they cannot be manipulated by an adversary).

This is different from satellite QKD, where the satellite itself might need to be trusted; in fibre BB84, the trust lies in the local devices.

Distance: Typically workable up to $\sim$ 80–100 km using good detectors and active stabilisation.

BBM92 In fibre:

- the entangled source is often placed in the middle to share the loss,
- time-bin entanglement is preferred for stability.

Trust assumptions: A key advantage of BBM92 is that the *source does not need to be trusted*. Even if the source is operated by an adversary, security can still be proven as long as the correlations between Alice and Bob match what quantum mechanics predicts.

However, Alice and Bob must still trust:

- their measurement devices,
- their basis choices,
- their random-number generators.

Distance: Typically tens of kilometres; can reach around 100 km with central placement of the source and high-performance detectors.

E91

E91 is the most conceptually powerful protocol, because security comes from violating a Bell inequality.

Why this matters: A Bell violation implies that:

- Alice and Bob share nonlocal correlations,
- no local realistic model can reproduce these correlations,
- an adversary cannot have full information about the outcomes.

Trust assumptions: In principle, E91 reduces trust because:

- the source can be fully untrusted,
- the correlations themselves certify security.

However:

- Bell tests are very demanding experimentally,
- fibre loss makes maintaining high visibility extremely difficult,
- loophole-free Bell violations usually require extremely low noise and high-efficiency detectors.

For this reason, E91 in fibre is possible (50–100 km demonstrations exist), but it is technically more challenging than BB84 or BBM92.

2.2.4 Operational Constraints in Fibre

Several practical issues affect how far and how reliably fibre QKD protocols can run:

- **Distance limits:** Exponential loss fundamentally restricts practical distances without repeaters.
- **Detector performance:** Long-distance fibre experiments rely on ultra-low-noise superconducting nanowire detectors.
- **Coexistence with classical traffic:** Raman noise increases the QBER and must be suppressed via filtering, spectral separation, and power control [17].
- **Stability requirements:** Interferometers for time-bin encoding require phase stabilisation at the kilohertz scale, and entanglement-based protocols require both temporal and spectral indistinguishability.

2.2.5 Security Considerations

Security proofs for fibre QKD are well established. In practice, the main concerns come from imperfections:

- **Detector side channels:** Particularly relevant for BB84. Entanglement-based protocols help reduce source-related vulnerabilities.
- **Source imperfections (in BB84):** Phase randomisation and intensity fluctuations affect security [19].
- **Raman noise and background counts:** These can mimic eavesdropping and increase the QBER [17].
- **Finite-key effects:** Long distances mean low count rates, which amplifies statistical fluctuations [20].

2.2.6 Future Directions

Future fibre QKD research focuses on increasing stability, distance, and practicality:

- **More stable interferometry:** Improved phase-stabilisation methods reduce drift and enable higher key rates.
- **Integrated photonics:** Chip-based phase modulators, interferometers, and entangled-photon sources improve robustness and reduce cost.
- **Improved coexistence with classical channels:** Better filtering and wavelength-division multiplexing help reduce Raman noise.
- **Higher-performance detectors:** Modern SNSPDs offer $> 90\%$ efficiency and extremely low dark counts, directly extending QKD distance.
- **Quantum repeaters (long-term):** Entanglement swapping and quantum memories represent the route to continental-scale E91/BBM92 networks, though still experimental [21].

2.2.7 Active Research Groups and Countries

Major contributors in fibre QKD research include:

- **China:** USTC and CAS work extensively on entanglement distribution and fibre QKD deployments [22].
- **Europe:** Geneva pioneered time-bin entanglement in fibre [18]. The UK Quantum Communications Hub deploys BB84 in metropolitan networks.
- **Japan:** NICT and Toshiba develop high-speed BB84 and entangled-photon sources at telecom wavelengths [23].
- **Canada:** Calgary and Waterloo groups focus on entanglement distribution and coexistence with classical traffic.

- **Singapore:** The Centre for Quantum Technologies works on chip-based QKD and integrated entangled sources [24].

2.3 Comparison of Fibre and Satellite QKD

Feature	Fibre QKD	Satellite QKD
Channel loss	Exponential with distance (0.2 dB/km).	Geometric-dominated; large absolute loss but not exponential with distance above atmosphere.
Operational range	Tens–hundreds of km (without repeaters).	Hundreds–thousands of km (single-hop intercontinental with satellites/relays).
Environmental stability	Very stable (no turbulence).	Highly dependent on weather, daylight and pointing.
Background noise	Raman scattering from co-propagating classical channels + detector dark counts.	Solar/sky background and atmospheric scattering (worse by day).
Encoding	Time-bin preferred; polarisation needs active compensation.	Polarisation natural/stable in vacuum; time-bin harder due to varying path lengths.
Hardware demands	Phase stabilisation, filtering, high-efficiency detectors (SNSPDs).	Precise pointing/tracking, adaptive optics, space-qualified optics/ground stations.
Trusted-node needs	BB84: trusted transmitter; BBM92: source can be untrusted; E91: trustless in principle but experimentally demanding.	BB84/BBM92 on satellite: satellite typically must be trusted; E91 could remove trust but is technically challenging.
Best-suited protocols	BB84 and BBM92 for practical deployments; E91 for short/controlled links.	BB84 and BBM92 for current use; E91/entanglement for future trust-free links.
Practical strengths	Mature, stable, easy network integration.	Global reach and ability to connect remote regions.
Main limitations	Rapid loss accumulation with distance.	Intermittent coverage, weather dependence, higher mission cost.

Table 1: At-a-glance comparison between fibre and satellite QKD (BB84, BBM92, E91).

When to Use Fibre vs. Satellite QKD

Fibre is the preferred medium for *metropolitan and regional* quantum networks: it is stable, available everywhere, and compatible with existing telecom infrastructure, though limited to a few hundred kilometres without quantum repeaters. Satellite QKD, in contrast, is the method of choice for *continental or global* key distribution, especially where laying fibre is impractical or where distances exceed the fibre loss limit. Satellites offer enormous reach but at the cost of intermittent operation, weather dependence, and the need to trust the satellite for prepare-and-measure protocols such as BB84 and BBM92. In principle, E91 could provide trust-free operation in both media, but practical loophole-free Bell tests remain challenging.

2.4 Security Proofs in Quantum Key Distribution

A *security proof* in QKD is a formal demonstration that the final secret key shared between two parties (Alice and Bob) is secure, even if an eavesdropper (Eve) has complete control over the communication channel. This means that Eve cannot gain meaningful information about the key without being detected.

To understand security proofs, it is useful to consider a few fundamental concepts: assumptions, attack models, how security is measured, and how real-world imperfections are handled.

2.4.1 Assumptions in Security Proofs

Even the strongest security proofs rely on some assumptions:

- **Trusted devices:** Alice's and Bob's devices behave as expected. They prepare and measure the states correctly. This is called a *device-dependent* model. Without this assumption, fully device-independent proofs are required, which are much more complex.
- **Authenticated classical channel:** Alice and Bob must be able to exchange classical messages without Eve impersonating them. This ensures that key reconciliation and error correction steps are not tampered with.
- **Worst-case eavesdropper:** Eve is allowed to perform any attack permitted by quantum physics. This includes intercepting, delaying, or modifying quantum signals, storing them in a quantum memory, and performing the best possible measurement later.
- **No assumptions about computational power:** Security is information-theoretic, meaning it does not rely on any computational hardness assumptions.

2.4.2 Trusted Nodes and Devices

Some parts of a QKD system must be trusted for the security proof to hold. This does not mean the entire network is vulnerable, but certain components cannot be assumed to be under the control of an adversary:

- **Sources:** The device that prepares quantum states (photons) must produce the states as expected. If the source is compromised, Eve might know the key without introducing detectable errors.
- **Detectors:** The measurement devices should not leak extra information. Imperfections like efficiency mismatch, timing leaks, or side channels can be exploited if not accounted for in the proof.
- **Intermediate nodes:** In networked QKD, nodes that forward or manage keys may need to be trusted unless entanglement-based protocols or device-independent schemes are used. Trust assumptions here define the network's security model.

2.4.3 Eavesdropper Models

Security proofs define what kind of attacks Eve can perform:

- **Intercept-resend attacks:** Eve measures photons and sends replacements. Such attacks introduce detectable errors.
- **Photon-number-splitting attacks:** If multi-photon pulses are emitted (common in real lasers), Eve can keep a photon and let the rest pass. Decoy-state methods help detect this.
- **Coherent attacks:** Eve interacts with multiple signals together, stores them in a quantum memory, and measures them optimally later. Modern proofs account for these general attacks.

2.4.4 Measuring Security

The key quantity in security proofs is the **Quantum Bit Error Rate (QBER)**:

$$\text{QBER} = \frac{\text{number of wrong bits}}{\text{total received bits}}. \tag{3}$$

Intuitively:

- Low QBER $\Rightarrow$ little disturbance, Eve has little information.
- High QBER $\Rightarrow$ significant disturbance, the key may be insecure.

Each protocol has a maximum tolerable QBER above which the protocol is aborted.

2.4.5 ε-Security: Quantifying the Risk

In modern QKD proofs, security is often expressed as *ε-security*. Here, ε is a small number representing the maximum probability that the final key is **not perfectly secure**:

- $\varepsilon_{\mathbf{sec}}$**:** Probability that Eve has some non-negligible information about the key.
- $\varepsilon_{\mathbf{cor}}$**:** Probability that Alice's and Bob's final keys are not exactly identical due to undetected errors.

Conceptually:

- If $\varepsilon_{\text{sec}} = 10^{-10}$, then there is only a one in ten billion chance that the key is insecure.
- These parameters determine how much *privacy amplification* is needed: the smaller ε, the more bits must be sacrificed to ensure security.

2.4.6 Privacy Amplification

After correcting errors, Eve may still have partial information about the key. Privacy amplification applies a hash function to compress the key, removing any information Eve could have. The amount of compression depends directly on ε_{sec}.

2.4.7 Handling Imperfections

Real devices are not perfect. Security proofs account for:

- **Source imperfections:** Multi-photon pulses, intensity fluctuations, or phase errors. Decoy states are used to estimate and mitigate multi-photon risks.
- **Detector imperfections:** Dark counts, efficiency mismatch, or side channels. These are included in the error budget to ensure secure key extraction.
- **Finite-key effects:** Real implementations use a limited number of signals. Statistical fluctuations are accounted for to guarantee security even with finite data.

2.4.8 Summary

A QKD security proof guarantees that:

- The final key is uniformly random and secret.
- Eve's information is negligible, even under the strongest quantum attacks.
- Security does not rely on assumptions about Eve's computational power.
- Device imperfections and finite data are explicitly included in the analysis.
- Certain nodes or devices must be trusted, and these assumptions are explicitly stated.
- The failure probability is quantified by ε, making the security guarantee precise and measurable.

2.4.9 Key Security Proof Literature

Below is a list of the main security proofs for BB84, BBM92, and E91:

BB84

- **Shor & Preskill (2000)**: *Simple Proof of Security of the BB84 Quantum Key Distribution Protocol.* Uses an entanglement-based reduction to CSS error-correcting codes. Asymptotic, unconditional security.
- **Renner (2005/2008)**: *Security of Quantum Key Distribution.* Introduces the modern composable security framework and smooth entropy tools. Finite-key and general security model.
- **Lo, Ma & Chen (2005)**: *Decoy State Quantum Key Distribution.* First full proof of decoy-state security against photon-number-splitting (PNS) attacks. Practical implementation security.

- **Tomamichel et al. (2012)**: *Tight Finite-Key Analysis for Quantum Cryptography.* Smooth min-/max-entropy framework giving near-optimal finite-key bounds.

BBM92

- **Mayers (1996, 2001)**: *Quantum Key Distribution and String Oblivious Transfer in Noisy Channels.* One of the first unconditional security proofs; entanglement-based formulation.
- **Lo & Chau (1999)**: *Unconditional Security of Quantum Key Distribution over Arbitrarily Long Distances.* Entanglement purification + error correction as the basis of security.
- **Renner et al. (2005)**: *Security of Quantum Key Distribution.* Finite-key, composable framework applied to entanglement-based protocols (including BBM92).
- **Ma, Fung & Lo (2007)**: *Quantum Key Distribution with Entangled Photon Sources.* Practical entangled-photon security analysis including losses and finite statistics.

E91

- **Ekert (1991)**: *Quantum Cryptography Based on Bell's Theorem.* Security from CHSH Bell inequality violation; no direct entropic proof.
- **Acín et al. (2007)**: *Device-Independent Security of Quantum Key Distribution.* First full device-independent (DI-QKD) proof based solely on Bell violation.
- **Renner et al. (2005)**: *Security of Quantum Key Distribution.* Composable, entropic security applied to entanglement-based schemes, including E91.
- **Pirandola et al. (2008, 2009)**: *Practical Security of Entanglement-Based QKD Over Lossy Channels.* Security analysis using Gaussian channel models and entanglement-based methods.

2.5 General Experimental Components

Quantum key distribution (QKD) systems rely on a chain of physical and classical processes that implement the abstract protocol in practice. These include the generation of quantum states, the physical encoding and measurement of qubits, their transmission through a quantum channel, the detection of extremely weak optical signals, and finally the classical post-processing procedures required to extract a secure key

2.5.1 Photon Sources

A QKD system requires a means of sending either single photons or entangled photon pairs. Since ideal deterministic single-photon sources are not widely available, practical implementations use two main technologies: weak coherent pulses (WCPs) and entangled photon pair sources based on spontaneous parametric down-conversion (SPDC).

Weak Coherent Pulses (WCP)

A weak coherent pulse is produced by heavily attenuating a laser so that the average photon number per pulse is well below one, typically denoted by μ. Lasers naturally emit light in a coherent state, which has a Poisson distribution of photon numbers. The probability of emitting n photons is

$$P(n) = \frac{\mu^n e^{-\mu}}{n!}.$$

For small μ (e.g., $\mu = 0.1$), most pulses contain either zero or one photon, with a small but nonzero probability of multi-photon pulses. These multi-photon components pose a security risk (they enable photon-number-splitting attacks), but the decoy-state method allows Alice and Bob to estimate the fraction of multi-photon pulses and prevent such attacks.

WCP sources are extremely simple and inexpensive since they consist of nothing more than a pulsed laser followed by strong optical attenuation. For this reason, they are the standard choice for prepare-and-measure protocols such as BB84.

Entangled Photon Sources (SPDC)

Entanglement-based protocols (such as BBM92 and E91) require pairs of photons exhibiting strong correlations. The most widely used entangled photon source is based on spontaneous parametric down-conversion (SPDC). A high-energy pump photon (usually from a UV laser) enters a nonlinear crystal such as BBO, KTP, or PPKTP. Inside the crystal, there is a tiny probability that one pump photon will spontaneously "split" into two lower-energy photons called the *signal* and *idler*. This process obeys strict conservation laws leading to strong correlations between the two photons.

If the SPDC setup is designed so that the pair can be produced in two indistinguishable ways (for example, horizontal–vertical or vertical–horizontal polarisation), the resulting output is an entangled state such as

$$\Psi^- = \frac{1}{\sqrt{2}}\big(H_s V_i - V_s H_i\big).$$

Depending on the arrangement, SPDC sources can create polarisation entanglement or time-bin (energy–time) entanglement. Time-bin entanglement is particularly stable in fibre systems, as it is immune to polarisation drift.

SPDC is probabilistic: pair creation is rare, so high-efficiency detectors and precise timing systems are required. Pulsed pump lasers help by providing a well-defined emission time, enabling accurate coincidence detection.

2.5.2 State Preparation and Measurement

To implement QKD protocols, Alice must encode quantum states, and Bob must measure them in randomly chosen bases. Although the theoretical description often uses polarisation or spin states, real implementations rely on optical devices that manipulate the phase, polarisation, or timing of photons.

State Preparation

In BB84, Alice prepares one of four states corresponding to two bases. This is accomplished using devices that modify the photon's polarisation or phase.

Polarisation encoding. In free-space systems, the most intuitive method is to encode qubits in the polarisation of the photon. Fixed waveplates (quarter-wave plates, half-wave plates) rotate the polarisation according to their optical axis, allowing Alice to generate H, V, D, and A states. More advanced setups use electro-optic modulators (EOMs), which act as voltage-controlled waveplates capable of very fast switching between states.

Phase and time-bin encoding. In optical fibre, polarisation drifts over time due to temperature and mechanical stress. To avoid this, many fibre QKD systems use phase encoding. Alice sends two pulses separated by a short time delay and modulates their relative phase using a phase modulator. This encodes the qubit in the phase difference between the pulses. In time-bin encoding, Alice creates two temporal bins ("early" and "late") using an interferometer and imposes a relative phase between them. This encoding is very stable in fibre and therefore widely used.

Measurement

Bob's measurement setup mirrors Alice's preparation. For polarisation encoding, Bob uses either passive beam splitters or active EOMs to select a basis. A polarising beam splitter then directs different polarisation components to different detectors.

For phase or time-bin encoding, Bob uses an interferometer matched to Alice's. When the early and late pulses interfere, the detection probabilities depend on their relative phase:

$$P_0 = \frac{1}{2}\big(1 + \cos\phi\big), \qquad P_1 = \frac{1}{2}\big(1 - \cos\phi\big),$$

where P_0 and P_1 are the probabilities of detection at the two output ports. The stability of Bob's interferometer is essential because any drift in its phase contributes directly to the quantum bit error rate (QBER).

2.5.3 Quantum Channels

The quantum channel is the physical medium through which photons propagate. In fibre, attenuation grows exponentially with distance. Although the exact fibre expression is discussed elsewhere, the essential effect is that only a fraction of photons reach Bob. Fibre also introduces polarisation drift, meaning that polarisation-based qubits degrade over long distances unless active compensation is applied.

Free-space and satellite channels experience losses determined mainly by diffraction, atmospheric turbulence, and pointing errors. These losses are often time-dependent, especially in satellite QKD where the link quality varies as the satellite moves across the sky. Background light (from the atmosphere or sunlight) also contributes noise, particularly in daytime operation.

2.5.4 Detection Systems

Because QKD signals are extremely weak, detection technologies must be capable of registering individual photons with high efficiency and minimal noise. Two common detector technologies are single-photon avalanche diodes (SPADs) and superconducting nanowire single-photon detectors (SNSPDs).

Single-Photon Avalanche Diodes (SPADs)

SPADs are semiconductor devices biased above their breakdown voltage. When a photon is absorbed, it creates an electron–hole pair that triggers an avalanche, producing a voltage pulse that indicates a detection event. After each detection, the device must be quenched briefly before it is ready again.

Although SPADs operate at or near room temperature and are inexpensive, they suffer from dark counts (false detections), afterpulsing, and relatively large timing uncertainty. These imperfections contribute to the QBER and reduce the secure key rate.

Superconducting Nanowire Single-Photon Detectors (SNSPDs)

SNSPDs are currently the best-performing detectors for QKD. A thin superconducting nanowire is cooled to cryogenic temperatures and biased with a current just below its critical value. When a photon is absorbed, a small region becomes resistive, producing a fast electrical pulse. SNSPDs combine high detection efficiency (up to 95%), extremely low dark counts, and excellent timing resolution. Their main drawback is the need for cryogenic operation.

2.5.5 Classical Post-Processing

Once detection is complete, Alice and Bob hold correlated but imperfect data. Several classical steps are required to produce a secure key.

Sifting. Alice and Bob reveal the bases they used and keep only the events where they matched.

Parameter estimation. They reveal a subset of their data to estimate the QBER and other relevant parameters. These estimated quantities determine the amount of privacy amplification needed later.

Error correction. Alice and Bob use classical algorithms (e.g., Cascade or LDPC codes) to correct discrepancies between their raw keys. The amount of information leaked during this stage must be tracked carefully.

Privacy amplification. Finally, Alice and Bob apply a universal hash function to compress their reconciled key into a shorter one that is secure even if an adversary has partial information. This ensures composable security:

$$\|\rho_{\text{real}} - \rho_{\text{ideal}}\|_1 \leq \varepsilon,$$

where ε is the overall failure probability of the protocol.

Authenticated classical communication ensures that an adversary cannot impersonate either party or modify the messages they exchange during these steps.

2.5.6 BB84 Procedure Overview

1. **Photon generation** — Weak coherent pulses produced by a pulsed laser and attenuator.
2. **State preparation** — Polarisation or time-bin encoding using modulators/interferometers.
3. **Decoy-state modulation** — Intensity switching between signal and decoy levels.
4. **Transmission** — Pulses sent through the quantum channel (fibre or free-space).
5. **Measurement** — Basis selection using polarisation optics or interferometers; time-tagged detection.
6. **Sifting** — Public basis revelation and retention of matched-basis events.
7. **Parameter estimation** — Use gains/QBER/decoy statistics to bound single-photon parameters.
8. **Error correction & privacy amplification** — Reconciliation and secure key extraction.

2.5.7 BBM92 Procedure Overview

1. **Entangled-pair generation** — Polarisation-entangled photon pairs produced by SPDC in a nonlinear crystal.
2. **Distribution of photons** — One photon sent to Alice, the other to Bob (via fibre or free-space).
3. **Basis selection** — Each party randomly chooses between two conjugate bases (e.g., Z, X) using passive or active modulators.
4. **Measurement** — Single-photon detectors record outcomes with time-tagging to identify entangled coincidences.
5. **Sifting** — Alice and Bob reveal their bases and keep only the events measured in the same basis.
6. **Parameter estimation** — Coincidence visibility and QBER are used to bound Eve's information.
7. **Error correction** — Classical reconciliation to align raw keys.
8. **Privacy amplification** — Compression of reconciled key to remove Eve's information and produce a final secure key.

2.5.8 E91 Procedure Overview

1. **Entangled-pair generation** — A source (trusted or untrusted) emits maximally entangled photon pairs via SPDC.
2. **Distribution of photons** — One photon sent to Alice and one to Bob through the quantum channel.
3. **Measurement-setting selection** — Alice and Bob each choose randomly among three measurement bases (e.g., A_1, A_2, A_3 and B_1, B_2, B_3).
4. **Bell-test measurements** — Detection events recorded with timestamps; used to evaluate CHSH correlators.
5. **Sifting** — A subset of outcomes is used for Bell-violation estimation; the compatible basis pair (e.g., A_1–B_1) forms the raw key.
6. **Bell-violation analysis** — CHSH value S estimated to bound Eve's information in a device-independent manner.
7. **Error correction** — Classical reconciliation to correct discrepancies in the raw key.
8. **Privacy amplification** — Secure final key extracted based on Bell-violation-derived entropy bounds.

Year	Experiment	Distance	Medium	Reported Rate	Group	Reference
1998	Early fibre entanglement QKD	∼0.4 km	Fibre	Coincidence rates: kHz-scale	Geneva (Gisin)	[18]
2004	50 km time-bin entangled QKD	50 km	Fibre	Sifted rate: few hundred bits/s	Geneva (Gisin)	[25]
2007	144 km free-space entangled QKD	144 km	Free-space	Coincidence rate: ∼1–10 Hz	Vienna IQOQI (Zeilinger)	[26]
2008	Real-time BBM92 over two free-space links	0.4–1.6 km	Free-space	Sifted key: ∼100–1000 bits/s	Geneva/Waterloo	[27]
2008	Entanglement over long-distance fibre	tens of km	Fibre	Coincidence rates: 10–100 Hz	NTT Japan	[28]
2013	300 km time-bin entanglement distribution	300 km	Fibre	Coincidence: ∼0.1–1 Hz	NTT Japan (Takesue)	[6]
2017	Satellite entanglement distribution (Micius)	1200 km	Satellite downlink	Coincidence: ∼1 pair/s	USTC (Pan)	[29]
2020	Satellite BBM92 with secure key generated	1120 km	Satellite downlink	**Secure key: ∼0.12 bits/s**	USTC (Pan)	[4]
2022	BBM92 over dusty free-space channel	200 m	Free-space	Secure + sifted rates reported (variable)	PRL India	[30]

2.6 Achievements

2.7 Scientific-review

2.7.1 Tittel et al. (1999): Quantum Cryptography using Energy–Time Bell States

Tittel *et al.* [18] present one of the earliest demonstrations of entanglement-based quantum key distribution (QKD) using discrete energy–time Bell states transmitted through optical fibre. Their work establishes the practical feasibility of distributing entanglement in real-world fibre links and using it to implement a BBM92-type protocol without requiring fast random basis selection. The authors generate photon pairs at 1310 nm (telecom O-band) via spontaneous parametric down-conversion (SPDC) in a $KNbO_3$ crystal pumped by a pulsed diode laser at 655 nm. The pump emits 600 ps (FWHM) pulses at 80 MHz, which are first passed through an unbalanced Michelson interferometer that introduces a temporal separation of $\Delta T \approx 1.2$ ns between two pump pulses. Provided that the probability for two-pair emission is kept well below 1%, the creation of a photon pair is described by the two-time-bin Bell state

$$|\psi\rangle = \frac{1}{\sqrt{2}}\Big(|s\rangle_A|s\rangle_B + e^{i\phi}|l\rangle_A|l\rangle_B\Big), \tag{4}$$

where $|s\rangle$ denotes pair creation initiated by a pump photon that travelled via the short arm of the pump interferometer and $|l\rangle$ by one that travelled via the long arm. The phase ϕ is controlled by the pump interferometer, and A, B denote Alice's and Bob's photons respectively.

Three time bins and measurement bases. Alice and Bob each employ an unbalanced fibre Michelson interferometer whose path-length difference matches that of the pump interferometer to within the coherence length of the downconverted photons ($\sim 20\,\mu$m). When detection times are recorded relative to the pump emission time t_0, each party observes photons in one of three non-overlapping time slots: (i) the left satellite peak ("early" events) corresponding to the processes $|s\rangle_P; |s\rangle_A$ or $|s\rangle_P; |s\rangle_B$; (ii) the central peak ("middle" events) arising from the indistinguishable alternatives $|s\rangle_P; |l\rangle_{A/B}$ and $|l\rangle_P; |s\rangle_{A/B}$; and (iii) the right satellite peak ("late" events) corresponding to $|l\rangle_P; |l\rangle_{A/B}$. The pump interferometer creates two well-separated pump pulses, so any SPDC pair is generated either at the "early" time t_0 or the "late" time $t_0 + \Delta T$. After passing through Alice's and Bob's equally unbalanced interferometers, each photon can again take either a short or a long path, leading to possible arrival times t_0, $t_0 + \Delta T$, and $t_0 + 2\Delta T$, which form the left, central, and right detection peaks. A left peak can only originate from the early pump pulse followed by two short paths ($|s\rangle_P; |s\rangle_A|s\rangle_B$), and a right peak only from the late pump pulse followed by two long paths ($|l\rangle_P; |l\rangle_A|l\rangle_B$). Mixed combinations such as "left at Alice, right at Bob" would require the *same* photon pair to have been created simultaneously by both the early and late pump pulses, which is impossible. Consequently, the arrival-time constraints forbid cross terms, ensuring that only perfectly correlated left–left and right–right events appear in the time basis in which Alice and Bob assign bit value 0 to short-arm processes and 1 to long-arm processes, yielding perfectly correlated bits.

The central peak realises the complementary *energy basis*. Here the two indistinguishable alternatives interfere, and the joint detection probabilities at Alice's and Bob's output ports $i, j = \pm 1$ (corresponding to "+" and "−" detectors) are given by

$$P_{i,j} = \frac{1}{2}\big(1 + ij\cos(\alpha + \beta - \phi)\big), \tag{5}$$

where α and β are the phases of Alice's and Bob's interferometers. By choosing appropriate phase settings, the two parties obtain fully correlated detection outcomes in the energy basis as well.

In analogy with BB84, the time basis supplies the key, while the energy basis provides the security check.

Experimental implementation. Behind the SPDC crystal, the downconverted photons are separated by a fibre coupler and transmitted to Alice and Bob through short fibre links. The interferometers use 3-port circulators and Faraday mirrors to stabilise polarisation. Detection is performed using passively quenched germanium avalanche photodiodes cooled to 77 K, operated with dark count rates of ~ 30 kHz and quantum efficiencies of $\sim 5\%$. Typical single-photon rates are 4–7 kHz, the variation arising from differing detector efficiencies and interferometer losses. Triple coincidences between the pump emission signal and detections at Alice and Bob are used to filter events into the appropriate time bins.

Results. In the time basis, correlated satellite-peak events occur at a rate of roughly 1700 coincidences per 100 s, while non-correlated events remain at the level of ~ 80 per 100 s. Over all detector pairings, the authors report quantum bit error rates (QBERs) between 3.7% and 5.7%, with a mean value of $(4.6 \pm 0.1)\%$. In the energy basis, interference fringes are observed in the central time bin with visibilities ranging from 89.3% to 94.5% and a mean visibility of $(92.2 \pm 0.8)\%$, corresponding to a violation of the Bell inequality by 27 standard deviations. For suitable phase settings, correlated central-peak events occur at a rate of ~ 800 per 50 s, yielding energy-basis QBERs between 2.8% and 5.4% with a mean of $(3.9 \pm 0.4)\%$.

The authors report an overall sifted key rate of approximately 33 Hz. They identify detector noise, imperfect temporal localisation of the pump pulse, finite timing resolution of the detectors, and non-perfect interferometric visibility as the dominant contributors to the non-zero QBER. Importantly, they note that increasing the total loss by 6 dB (equivalent to roughly 20 km of optical fibre under ideal conditions) increases the QBER by only about 1%, since detector-noise-induced errors remain nearly constant with distance.

2.7.2 Marcikic et al. (2004): Time-bin entanglement and QKD over 50 km of optical fibre

Marcikic *et al.* [25] represent the next major milestone after the early 1998 Geneva experiments. The motivation is no longer simply to demonstrate energy–time entanglement, but to show that time-bin qubits can be distributed over distances relevant for real optical networks, while maintaining sufficient phase stability to violate a CHSH Bell inequality and to perform proof-of-principle QKD. The entire experiment is carried out at telecommunication wavelengths (1.3 µm and 1.55 µm), allowing photon transmission through 50 km of fibre with practical losses.

Source and state preparation

As in the 1998 experiment, time-bin entanglement is created using a pump interferometer followed by SPDC. However, the implementation is significantly more advanced. A 150 fs, 710 nm pulsed laser passes through a bulk Michelson interferometer with a path-length imbalance of $\Delta\tau = 1.2$ ns, producing two pump time bins. Downconversion in an LBO crystal generates non-degenerate photon pairs at 1.3 µm and 1.55 µm, which are separated using a wavelength-division multiplexer and sent through 25.3 km of standard fibre (Alice) and 25.3 km of dispersion-shifted fibre (Bob). The resulting entangled state is

$$|\Psi\rangle = \frac{1}{\sqrt{2}}\big(|0\rangle_A|0\rangle_B - e^{i\phi}|1\rangle_A|1\rangle_B\big),$$

where $|0\rangle$ and $|1\rangle$ denote the early and late time bins, and the phase ϕ is defined relative to the pump interferometer.

Compared to the 1998 work, this experiment operates in a regime where chromatic dispersion, multi-pair emission, and environmental drifts are far more significant. Dispersion-compensating fibre and spectral filtering are therefore required to prevent overlap between time bins and to suppress multi-photon contributions.

Actively stabilised long interferometers

A central advance of the work is the *active* phase stabilisation of *all three* interferometers (pump, Alice, Bob). A frequency-stabilised 1534 nm laser probes each interferometer, and feedback is applied via piezoelectric actuators. The stabilisation is periodically interleaved with short measurement windows (100 s measurement followed by 5 s re-locking). This enables stable phase control over several hours—a requirement for long-distance Bell tests and entanglement-based QKD.

This is a major improvement over earlier demonstrations. The 1998 interferometers were laboratory-scale and passively stable; here the fibre lengths and environmental sensitivity necessitate active feedback.

Two-photon interference and Bell inequality violation

Time-bin qubits are analysed via the central-peak interference described in the previous subsection. When both Alice and Bob detect in the central peak, the coincidence rate between detectors A_iB_j obeys

$$R_{A_iB_j}(\alpha, \beta, \phi) \propto 1 + ij\,V\cos(\alpha + \beta - \phi),$$

with $i, j = \pm 1$ and visibility V. The corresponding correlation coefficient is

$$E(\alpha, \beta) = V \cos(\alpha + \beta).$$

Using phase settings appropriate for the CHSH inequality, they obtain a raw visibility of 78%, yielding

$$S = 2.206 \pm 0.045,$$

a violation by more than four standard deviations without subtracting detector noise. Using a full four-setting measurement according to the textbook CHSH protocol, they report

$$S = 2.185 \pm 0.006,$$

which corresponds to a violation by more than 15 standard deviations—the strongest long-distance time-bin Bell test at the time.

In contrast, Tittel *et al.* (1998) achieved high visibility but only over sub-kilometre fibre, with no active stabilisation and no long-distance Bell measurement. The 2004 experiment thus marks the transition to *practical*, telecom-compatible entanglement distribution.

Proof-of-principle QKD over 50 km

The authors then implement an entanglement-based BBM92 protocol using the two complementary bases introduced previously:

- **Time basis (satellite peaks):** projects onto $|0\rangle$ or $|1\rangle$, used for key generation.
- **Energy basis (central peak):** projects onto $\frac{1}{\sqrt{2}}(|0\rangle \pm e^{i\alpha}|1\rangle)$, used for security testing.

Alice uses her detection time to trigger Bob's gated InGaAs APDs. In the time basis, the raw key rate is ~ 5 Hz with a measured QBER of 12.8%. In the energy basis, the rate is ~ 6 Hz with a QBER of 10.5%. The dominant error sources are accidental coincidences (detector dark counts), multi-pair emissions (8% probability at the given pump power), and small interferometer misalignments.

The average QBER of 11.5% lies below the 15% threshold for security against individual attacks, demonstrating that entanglement-based QKD over 50 km fibre is feasible with time-bin encoding.

This work therefore establishes time-bin encoding as a robust and practical method for long-distance entanglement distribution, paving the way for later experiments over hundreds of kilometres and for satellite-based implementations.

2.7.3 Ursin et al. (2007): Entanglement Distribution over a 144 km Free-Space Channel

Ursin *et al.* [26] report the first long-distance free-space distribution of polarization-entangled photons over a 144 km atmospheric channel between La Palma and Tenerife. This experiment established the feasibility of satellite-to-ground quantum communication by demonstrating that entanglement survives propagation through a turbulent, high-loss, unbounded optical path. Unlike fibre-based time-bin QKD, this work relies on polarization-entangled qubits and must contend with diffraction, pointing instability, atmospheric turbulence, and a time-varying link transmissivity. **Source and state preparation**

Polarization-entangled photon pairs are generated via type-II SPDC in a beta-barium borate (BBO) crystal pumped by a continuous-wave 405 nm diode laser. After spectral filtering and walk-off compensation, the prepared state ideally corresponds to the Bell state

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}\big(|H\rangle_A|V\rangle_B - |V\rangle_A|H\rangle_B\big),$$

where H and V denote horizontal and vertical polarization. One photon is measured locally on La Palma, while the partner photon is transmitted across the 144 km free-space link to the receiving station on Tenerife.

Free-space transmission system

The transmitted photon is expanded to a 15 cm beam diameter using a refractive telescope in order to reduce diffraction. Over the 144 km path, the beam experiences severe broadening due to diffraction, atmospheric turbulence, and beam wander, resulting in an expected channel loss of 30–40 dB under favourable conditions and up to 50 dB during periods of strong turbulence. Bidirectional beacon lasers at 532 nm are used for continuous pointing and tracking, enabling the transmitter and receiver to maintain microradian-level alignment. The receiver on Tenerife is the ESA Optical Ground Station (OGS), a 1 m telescope originally designed for optical satellite communication.

Model and channel characterisation

Unlike guided fibre channels, the free-space link exhibits time-varying, stochastic transmission determined by atmospheric turbulence. The detected coincidence rate can be modelled as

$$C(t) = R_0\, \eta_{\mathrm{atm}}(t)\, \eta_{\mathrm{opt}}\, \eta_{\mathrm{det}},$$

where R_0 is the source pair rate, η_{opt} the optical efficiency of the telescopes and coupling optics, η_{det} the detector efficiency, and $\eta_{\mathrm{atm}}(t)$ the atmospheric transmissivity, which fluctuates on millisecond to second timescales. The latter is influenced by:

- **Beam wander**: low-frequency turbulence causes the beam centroid to drift, periodically reducing telescope coupling.
- **Scintillation**: rapid intensity fluctuations described by a log-normal or gamma-gamma distribution.
- **Diffraction broadening**: causing irreversible geometric loss.

The total channel transmission often drops below 10^{-4}, meaning that background photons and detector dark counts can appreciably affect the signal-to-noise ratio (SNR). The accidental coincidence rate is modelled as

$$A = S_A S_B\, \tau_c,$$

where S_A and S_B are the single count rates at Alice and Bob, and τ_c is the coincidence timing window (typically a few nanoseconds). A high SNR is required to observe Bell inequality violations; thus narrow spectral filters and precise temporal gating are essential.

Polarization analysis and detection

At the receiver, the collected photons are analysed using half-wave plates and polarizing beam splitters to implement the required polarization measurement settings. Silicon avalanche photodiodes with narrow interference filters are used to minimize background noise. Polarization compensation is achieved by sending a reference laser through the same optical path and adjusting the analyser angles to correct any rotation induced by the telescope optics or misalignment.

Bell inequality violation

Polarization correlations are measured in four settings appropriate for a CHSH test. The correlation coefficient is computed from the coincidence counts C_{ij} corresponding to detector outcomes $i, j \in \{+, -\}$:

$$E(\alpha, \beta) = \frac{C_{++} + C_{--} - C_{+-} - C_{-+}}{C_{++} + C_{--} + C_{+-} + C_{-+}}.$$

Using these correlations, the Bell parameter

$$S = E(a, b) - E(a, b') + E(a', b) + E(a', b')$$

is obtained. Despite strong atmospheric fluctuations and channel losses exceeding 35 dB, the measured result

$$S = 2.508 \pm 0.037$$

violates the CHSH inequality by more than 13 standard deviations, confirming that entanglement survives transmission through the free-space channel.

2.7.4 Erven et al. (2008): Real-Time Entanglement-Based QKD over Urban Free-Space Links

Erven *et al.* [27] report the first real-time implementation of entanglement-based BBM92 QKD over two independent free-space optical links in an urban environment, with link lengths ranging from 0.4 to 1.6 km. In contrast to long-distance Bell tests, the focus here is on autonomous key generation, continuous tracking and synchronization, real-time sifting, and complete error correction and privacy amplification. Both photons of each entangled pair traverse their own atmospheric channels, making the setup an important precursor to practical urban free-space quantum networks.

Source and state preparation

The entanglement source is a Sagnac-loop type-II SPDC system pumped at 405 nm, generating polarization-entangled photon pairs close to the Bell state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}\left(|H\rangle_A|H\rangle_B + |V\rangle_A|V\rangle_B\right).$$

One photon is sent to Alice over the first free-space link, while the partner photon is sent to Bob over the second link. The Sagnac geometry provides high phase stability, high spectral purity, and symmetric spatial modes suitable for long-term autonomous operation.

Free-space transmission and tracking

Each link uses a beam-expanding telescope, a narrow field-of-view receiving telescope, and a bidirectional beacon laser for closed-loop pointing control. Atmospheric turbulence, building motion, and beam wander introduce time-varying losses of 4–12 dB depending on distance and weather. Both channels are subject to independent turbulence, and total coincidence rates therefore scale with the *product* of the two link transmissions.

Measurement and basis choice

BBM92 is implemented using passive basis selection: a 50/50 beam splitter routes each incoming photon into either the H/V or D/A ($\pm 45°$) polarization analyser. Silicon APDs detect the photons, and events are time-tagged with sub-nanosecond precision. Alice transmits her timestamps to Bob over a classical wireless network for real-time coincidence identification and sifting.

Model and key-rate analysis

Since both photons traverse lossy atmospheric channels, the expected coincidence rate is

$$C = R_0\, \eta_A \eta_B\, \eta_{\text{atm}}^{(A)} \eta_{\text{atm}}^{(B)},$$

where R_0 is the SPDC pair rate, $\eta_{A,B}$ are the optical and detection efficiencies at Alice and Bob, and $\eta_{\text{atm}}^{(A,B)}$ are the time-varying atmospheric transmissions of the two links.

The quantum bit error rate arises from accidental coincidences, background photons, and polarization misalignment:

$$\text{QBER} = \frac{E_{\text{acc}} + E_{\text{bg}} + E_{\text{pol}}}{S_{\text{tot}}}.$$

Accidental coincidences follow

$$A = S_A S_B\, \tau_c,$$

where S_A and S_B are the singles rates and τ_c is the coincidence window.

After sifting, Alice and Bob perform error correction and privacy amplification. The secret key fraction for BBM92 under individual attacks is estimated as

$$r = 1 - f\, h_2(\text{QBER}) - h_2(\text{QBER}),$$

where f is the error-correction inefficiency ($f \approx 1.15$) and h_2 the binary entropy. The final secure key rate is therefore

$$R_{\text{sec}} = R_{\text{sift}}\, r.$$

Experimental results

Table 2 summarises the key performance metrics for link distances between 0.41 and 1.62 km, including sifted and secure key rates as reported in [27].

Distance (km)	Singles (kHz)	Coinc. (Hz)	QBER (%)	Sifted Rate (bits/s)	Secure Rate (bits/s)
0.41	150	10.5	4.8	950	565
0.53	130	8.7	5.8	780	420
1.04	90	4.9	6.9	470	210
1.32	75	3.6	7.5	350	150
1.62	55	2.1	9.3	190	40

Table 2: Performance of the entanglement-based QKD system at different free-space distances, summarising typical singles rates, coincidence rates, QBER, sifted key rates, and final secure key rates as reported in [27].

2.7.5 Honjo et al. (2008): Long-Distance Distribution of Time-Bin Entanglement over Telecom Fibre

Honjo *et al.* [28] report the distribution of time-bin entangled photon pairs over long spans of optical fibre using a high-brightness periodically poled lithium niobate (PPLN) waveguide source. This work demonstrates that high-visibility Franson-type two-photon interference can be maintained after transmission over tens of kilometres of standard telecom fibre, establishing a practical foundation for long-distance entanglement-based quantum communication in the C-band.

Source and state preparation

Photon pairs at 1542 nm are generated by type-0 SPDC in a PPLN waveguide pumped by a continuous-wave 771 nm laser. The pump passes through an unbalanced Michelson interferometer, producing two time bins separated by $\Delta T \approx 1.6$ ns, thereby preparing the energy–time entangled state

$$|\psi\rangle = \frac{1}{\sqrt{2}}\big(|s\rangle_A|s\rangle_B + e^{i\phi}|l\rangle_A|l\rangle_B\big),$$

where $|s\rangle$ and $|l\rangle$ denote early and late time bins. The photon pairs are separated at a fibre coupler and sent to Alice and Bob through standard telecom fibre. Because the pump is continuous, the coherence time of the downconverted photons is determined primarily by spectral filtering.

Fibre transmission and dispersion management

Transmission over long fibre distances requires controlling chromatic dispersion and maintaining the indistinguishability of the two long–short alternatives in the Franson interferometer. The experiment employs dispersion-shifted fibre (DSF) near 1550 nm and narrowband filters of width 0.1 nm to increase the photon coherence time. Alice's and Bob's analysing interferometers are temperature-stabilised and actively phase-controlled to maintain path-length matching within the coherence length of the filtered photons. These measures ensure that Franson interference remains observable after tens of kilometres of transmission.

Detection and stabilisation

The analysing interferometers at Alice and Bob have the same path imbalance as the pump interferometer, enabling the observation of the three time bins characteristic of Franson interferometry. InGaAs avalanche photodiodes operated in gated mode provide typical detection efficiencies of 10–15% and dark count rates of a few kilohertz. Interferometer phases are controlled using piezoelectric actuators, allowing stable visibility measurements over extended periods.

Model and performance characterisation

The coincidence rate after transmission through a fibre of loss L (in dB) is

$$C = R_0\,\eta_A\eta_B\,10^{-L/10},$$

where R_0 is the pair generation rate and η_A, η_B are the total detection efficiencies at Alice and Bob. The observed interference visibility is defined as

$$V = \frac{C_{\max} - C_{\min}}{C_{\max} + C_{\min}},$$

and decreases with fibre length due to chromatic dispersion, multi-pair emissions, detector timing jitter, and residual path mismatch. The requirements for observing Franson interference are

$$|\Delta T_A - \Delta T_B| \ll \tau_c, \qquad \Delta T \gg \tau_{\text{coh}},$$

where τ_c is the two-photon coherence time and τ_{coh} the single-photon coherence time. The use of DSF and narrowband filtering ensures these constraints are satisfied over long fibre spans.

Experimental results

The experiment distributes time-bin entanglement over distances up to 60 km, maintaining high-visibility interference. Table 3 summarises representative visibilities and coincidence rates.

Distance (km)	**Visibility** (%)	**Coincidence Rate** (Hz)
0	93–95	~ 100
20	90–93	~ 50
40	88–92	~ 20
60	85–90	~ 10

Table 3: Representative two-photon interference visibilities and coincidence rates for different fibre lengths, based on the measurements reported in [28].

2.7.6 Inagaki et al. (2013): 300 km Distribution of Time-Bin Entanglement in Optical Fibre

Inagaki *et al.* [6] report the first demonstration of time-bin entanglement distribution over 300 km of optical fibre. This experiment extends the earlier long-distance fibre work of Honjo *et al.* [28] by combining ultra-narrowband filtering, dispersion-shifted fibre (DSF), active interferometer stabilisation, and superconducting single-photon detectors (SSPDs). The result is the longest fibre-based entanglement distribution achieved without quantum repeaters.

Source and state preparation

Photon pairs at 1550 nm are generated via SPDC in a high-brightness PPLN waveguide, pumped by a continuous-wave 775 nm laser. The pump passes through an unbalanced interferometer, creating two pump time bins with separation $\Delta T \approx 1.6$ ns and preparing the standard energy–time entangled state

$$|\psi\rangle = \tfrac{1}{\sqrt{2}}\big(|s\rangle_A|s\rangle_B + e^{i\phi}|l\rangle_A|l\rangle_B\big)$$

(as defined previously in the Honjo *et al.* subsubsection). A key advance is the use of extremely narrow 100 MHz filters, yielding a photon coherence time of $\tau_c \sim 10$ ns, over an order of magnitude larger than in earlier fibre-based experiments. This extended coherence time is essential to tolerate dispersion over 300 km.

Ultra-long fibre transmission and dispersion management

Transmission is performed using 300 km of dispersion-shifted fibre with total attenuation of ~ 60 dB. As in Honjo *et al.*, the requirements for Franson interference are

$$|\Delta T_A - \Delta T_B| \ll \tau_c, \qquad \Delta T \gg \tau_{\text{coh}},$$

but the greatly increased photon coherence time makes these constraints achievable at ultra-long distances.

Both analysing interferometers are actively phase-locked using a classical reference laser. Temperature control to the millikelvin level prevents slow drifts over the multi-hour measurement periods required at sub-hertz coincidence rates.

Detection using superconducting nanowire detectors

A central enabling technology is the use of superconducting single-photon detectors (SSPDs), which offer quantum efficiencies of 55–60%, timing jitter of ~ 65 ps, and dark count rates of ~ 100 Hz. Such performance is required because coincidence rates after 300 km are on the order of 0.1–1 Hz, far below the practical limits of InGaAs APDs.

Model and performance characterisation

The coincidence rate follows the standard attenuation model described in the previous fibre subsubsections:

$$C(L) = R_0\, \eta_A \eta_B\, 10^{-L/10},$$

with $L \approx 60$ dB for 300 km. Two-photon interference is quantified by the usual Franson visibility (see Tittel and Honjo subsubsections):

$$V = \frac{C_{\max} - C_{\min}}{C_{\max} + C_{\min}}.$$

At ultra-long distances, the dominant limitations are multipair emissions and accidental coincidences rather than chromatic dispersion. Accidentals follow the standard form

$$A = S_A S_B\, \tau_c + S_A D_B \tau_c + D_A S_B \tau_c,$$

where $S_{A,B}$ are singles and $D_{A,B}$ the dark-count rates. Because τ_c is large (tens of nanoseconds), narrow detection windows and low dark counts are essential.

Experimental results

Inagaki *et al.* distribute entanglement over 50–300 km. Representative visibilities and coincidence rates are summarised in Table 4. Even at 300 km, the visibility remains above the entanglement threshold.

Distance (km)	**Visibility** (%)	**Coincidence Rate** (Hz)
50	~ 90	~ 10
100	~ 88	~ 5
200	~ 80	~ 1
300	76.2 ± 3.6	0.1–1

Table 4: Two-photon interference visibility and coincidence rate at various fibre lengths, based on Inagaki *et al.* [6].

A Bell inequality is violated after background subtraction:

$$S = 2.19 \pm 0.10,$$

confirming the survival of entanglement after 300 km of fibre.

2.7.7 Yin et al. (2017): Satellite-Based Entanglement Distribution over 1200 km

Yin *et al.* [29] report the first successful distribution of polarization entanglement from a satellite to two widely separated ground stations. Using the *Micius* quantum science satellite in a 500 km sun-synchronous orbit, the experiment delivers entangled photon pairs to the Delingha and Lijiang ground stations, separated by 1203 km, establishing the longest-distance entanglement distribution demonstrated to date.

Satellite payload and entanglement source

The satellite hosts a high-brightness Sagnac-loop SPDC source based on type-II PPKTP, pumped by a 405 nm diode laser. It generates photon pairs entangled in polarization, ideally in the singlet state

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}\big(|H\rangle_A|V\rangle_B - |V\rangle_A|H\rangle_B\big).$$

The source brightness reaches $\sim 5.9 \times 10^6$ pairs/s, with the photons filtered and coupled into two independent optical downlinks. The optical assembly includes a 300 mm telescope, multi-stage vibration isolation, and active thermal control to maintain high source stability during orbital operation.

Space-to-ground transmission and pointing

Each photon of the entangled pair is directed to a separate ground station using independent telescope assemblies. As in previous free-space work (e.g. Ursin *et al.*), the main challenges are optical diffraction, atmospheric turbulence, and pointing stability. These are mitigated using a two-stage tracking system:

coarse pointing via a gimbal mechanism and fine pointing via a fast-steering mirror locked to ground-based beacon lasers. The total downlink loss varies between 40 dB and 60 dB depending on satellite elevation, dominated by geometric spreading and pointing errors rather than atmospheric absorption.

Both ground stations employ metre-scale telescopes, narrow spectral filters, and single-mode fibre coupling to suppress background light. Detection is performed using silicon avalanche photodiodes operated in low-noise conditions. Optical beacons and GPS-disciplined clocks synchronise the time-tagging systems at the two stations, allowing coincidence windows well below a nanosecond.

Channel model and performance characterisation

The downlink efficiency to each station is determined by diffraction, pointing accuracy, and atmospheric transmission. The overall two-photon coincidence rate may be written as

$$C(t) = R_0\, \eta_A(t)\, \eta_B(t)\, \eta_{\mathrm{det}}^2,$$

where R_0 is the pair generation rate at the satellite, $\eta_{A,B}(t)$ are the time-dependent channel efficiencies for each downlink, and η_{det} the detector efficiency. The atmospheric contribution to loss is small compared to diffraction and pointing, since only ~ 10 km of atmosphere is traversed vertically.

During favourable orbital passes, the downlink yields singles at the kHz level and coincidence rates around 1 Hz. Despite these low rates, the signal-to-noise ratio remains sufficient for entanglement verification because of strong spatial, spectral, and temporal filtering.

Experimental results

Polarisation correlation measurements in multiple bases yield a clear violation of the CHSH Bell inequality. The measured Bell parameter is

$$S = 2.37 \pm 0.09,$$

exceeding the classical bound by more than four standard deviations. This constitutes a decisive demonstration that high-fidelity entanglement can be distributed from space to widely separated locations on Earth. The experiment thus confirms that satellite-based quantum links can outperform fibre-based distribution at global scales, where optical fibre attenuation would prohibit any useful entanglement transmission.

2.7.8 Yin et al. (2020): Satellite-Based BBM92 QKD with Secure Key Generation over 1120 km

Building directly on the satellite-to-ground entanglement distribution reported in Yin *et al.* (2017), the experiment of Yin *et al.* (2020) demonstrates the first operational satellite-based entanglement QKD with positive secure key generation. While the 2017 study focused on verifying entanglement through Bell-inequality violations, the 2020 experiment performs a full implementation of the BBM92 protocol between two ground stations separated by 1120 km, including sifting, QBER estimation, error correction, and privacy amplification.

Experimental configuration

The satellite payload and optical downlinks are essentially identical to those used in the 2017 experiment: a high-brightness Sagnac-loop PPKTP SPDC source, dual 300 mm satellite transmitters, and metre-scale receiving telescopes on the ground with coarse and fine tracking. Because these elements have already been described in the previous subsubsection, we only emphasise the modifications required for QKD:

- Both ground stations implement random basis selection in the Z and X bases for BBM92 (polarisation analysis).
- A high-throughput time-tagging and synchronisation system enables real-time coincidence identification under rapidly varying link loss.
- Dedicated QKD-processing hardware performs sifting, error correction, and privacy amplification during each satellite pass.

Compared to the 2017 experiment, the system must maintain not only high entanglement fidelity but also sufficiently low QBER and adequate coincidence rates for extracting secret key, making the technical requirements significantly more stringent.

Channel characteristics

The downlink losses are of the same order as in Yin *et al.* (2017), varying between 40 dB and 60 dB depending on elevation angle and residual pointing errors. Coincidence rates reach the hertz level during favourable periods of the satellite pass. As before, diffraction and pointing dominate the loss budget, while atmospheric absorption plays only a minor role.

The coincidence rate follows the standard expression already used in the 2017 subsubsection,

$$C(t) = R_0\, \eta_A(t)\, \eta_B(t)\, \eta_{\mathrm{det}}^2,$$

but here the emphasis lies on maintaining a sufficiently high ratio of true to accidental coincidences to keep QBER within the acceptable QKD range. Typical QBERs lie between 1% and 3%, considerably lower than would be feasible in a terrestrial free-space link of comparable length.

BBM92 key extraction

Matched-basis measurement outcomes form the raw key. The secure key length after error correction and privacy amplification is

$$N_{\mathrm{secure}} = N_{\mathrm{raw}}\big[1 - h_2(Q)\big] - N_{\mathrm{leak}} - N_{\mathrm{safety}},$$

where h_2 is the binary entropy function and Q is the QBER. This is the same security expression used in standard entanglement-based QKD and matches the formulation introduced earlier in this review.

Using LDPC-based error correction and blockwise privacy amplification, the experiment obtains sifted key rates of order 0.5 bits/s and a final secure key rate of approximately 0.12 bits/s.

2.7.9 Mishra et al. (2022): BBM92 QKD over a Dusty Free-Space Channel

Mishra *et al.* (2022) investigate entanglement distribution and BBM92 quantum key distribution over a short but environmentally challenging free-space link. Unlike earlier long-distance free-space demonstrations such as Ursin *et al.* (2007), which operated in clear atmospheric conditions over 144 km, this study focuses on the effects of airborne dust and particulates on a $\sim$ 200 m terrestrial optical channel. The goal is to understand how real-world pollution and scattering conditions impact entanglement visibility, QBER, and the viability of practical urban QKD links.

Entanglement source and state preparation

The experiment employs a type-II SPDC source producing photon pairs entangled in polarisation. After spectral filtering and fibre coupling, the pairs are distributed to two free-space terminals forming a line-of-sight link across an outdoor environment with controlled or naturally varying dust concentration. The prepared state is the familiar singlet

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}\big(|H\rangle_A|V\rangle_B - |V\rangle_A|H\rangle_B\big),$$

allowing direct implementation of the BBM92 protocol through measurements in the Z and X bases. Because polarisation states are highly sensitive to scattering and depolarisation, the channel provides a stringent test of entanglement robustness in adverse atmospheric conditions.

Dusty free-space channel and degradation mechanisms

Transmission occurs through an outdoor path containing suspended dust particles. In contrast with long-distance free-space or satellite downlinks, the dominant channel impairments are Mie scattering, particulate absorption, and turbulence-induced coupling fluctuations. These processes reduce the channel efficiency η_{ch}, introduce depolarisation, and increase the ratio of accidentals to true coincidences.

The detected coincidence rate follows the standard form

$$C = R_0\, \eta_{\mathrm{ch}}\, \eta_A \eta_B,$$

while accidental coincidences are well described by

$$A = S_A S_B\, \tau_c,$$

as used in earlier free-space and satellite subsubsections. Increasing dust concentration reduces S_A, S_B, and C, but leaves A comparatively less suppressed, leading to higher QBER and reduced visibility. The depolarising effect of scattering can be viewed as a channel map $\rho \to \Lambda(\rho)$ which diminishes the two-photon interference visibility

$$V = \frac{C_{\max} - C_{\min}}{C_{\max} + C_{\min}}.$$

BBM92 key extraction

After basis selection and sifting, the raw key is processed using classical error correction and privacy amplification. The final secure key length is computed using the standard expression already employed in previous subsubsections:

$$N_{\text{secure}} = N_{\text{raw}}\big[1 - h_2(Q)\big] - N_{\text{leak}} - N_{\text{safety}},$$

where Q is the measured QBER. As dust concentration increases, visibility decreases and Q rises, eventually surpassing the threshold above which no positive secret key can be extracted.

Results

At low dust levels, the experiment reports high visibility ($> 90\%$) and QBER in the secure regime, enabling positive secret-key rates. As particulate density increases, scattering produces significant depolarisation, reducing visibility to below $\sim 70\%$ and causing QBER to reach values (5–15%) that limit or eliminate secure key generation.

This study demonstrates that secure entanglement-based QKD is feasible in dusty urban environments, but only within a limited range of particulate densities. Because many practical free-space quantum links—for example, last-mile access points, rooftop connections, or intra-campus links—operate in such conditions, these results are essential for realistic modelling and deployment of terrestrial free-space QKD systems.

2.7.10 Bourgoin et al. (2013): Comprehensive Modelling of LEO Satellite Quantum Communication

Bourgoin *et al.* (2013) provide a detailed modelling framework for quantum communication between low-Earth orbit satellites and ground stations. Unlike experimental satellite demonstrations such as Yin *et al.* (2017, 2020), this work develops a complete link-budget model that decomposes the satellite-to-ground optical channel into physical contributions and predicts the time-dependent transmissivity, background counts, and achievable key rates over entire satellite passes.

Optical link model

The total channel transmissivity is expressed as

$$\eta_{\text{ch}}(t) = \eta_{\text{geo}}(t)\,\eta_{\text{atm}}(t)\,\eta_{\text{point}}(t)\,\eta_{\text{opt}},$$

where the four terms correspond to geometric diffraction loss, atmospheric transmission, pointing error, and internal optical loss, respectively. This decomposition isolates the dominant loss mechanisms that vary as the satellite moves through its orbit.

The geometric term is derived from Gaussian-beam propagation. For a transmitter with divergence θ and a receiver with aperture D_{rx}, the captured power fraction is

$$\eta_{\text{geo}}(t) = \left(\frac{D_{\text{rx}}}{2w(t)}\right)^2,$$

with $w(t)$ the beam radius at the receiver plane. The slant range $L(t)$, which varies significantly during a LEO pass, directly affects $w(t)$ and therefore $\eta_{\text{geo}}(t)$.

Atmospheric transmission $\eta_{\text{atm}}(t)$ is obtained from zenith-dependent attenuation data and accounts for molecular absorption and scattering. Pointing loss $\eta_{\text{point}}(t)$ is modelled using a Gaussian distribution of residual pointing error from the satellite's tracking system.

Background noise

The model includes both sky radiance and detector dark counts, giving a total background

$$B = B_{\text{sky}} + B_{\text{dark}}.$$

The sky contribution strongly depends on satellite elevation, wavelength, moonlight, and observation conditions. This is crucial because the low transmissivity of satellite links makes entangled-photon detection susceptible to accidental coincidences.

Predicted performance

Using the above model, Bourgoin *et al.* compute the instantaneous coincidence rate

$$C(t) = R_0\, \eta_{\text{ch}}(t)\, \eta_{\text{det}}^2,$$

and integrate it over the visible portion of the pass to obtain the total number of detected pairs, sifted key bits, and secret key bits. The time dependence is important: the channel is weakest at low elevation angles and strongest near the zenith.

The study finds that:

- Diffraction and pointing loss dominate the overall channel attenuation.
- Atmospheric attenuation is comparatively moderate at optical wavelengths except at very low elevations.
- Only a portion of a LEO pass yields secret key because QBER increases sharply at low elevation angles.
- Increasing transmitter aperture or improving pointing precision significantly enhances key rates.

2.7.11 Khatri et al. (2018): Global-Scale Free-Space Entanglement Distribution

Khatri *et al.* (2018) develop an analytical model of global free-space entanglement distribution using a bosonic-channel framework. While earlier studies such as Bourgoin *et al.* model the optical link in terms of loss components, this work examines how attenuation and background noise directly affect entanglement measures such as negativity. It therefore complements link-budget analyses by quantifying how much entanglement survives extremely long free-space paths.

Channel model and transmissivity

The satellite downlink is treated as a pure-loss channel with transmissivity

$$\eta(t) = \eta_{\text{atm}}(t) \left(\frac{D_{\text{rx}}}{2L(t)\theta} \right)^2,$$

where $L(t)$ is the slant range, θ the beam divergence, and D_{rx} the receiver aperture. This expression resembles the geometric loss in Bourgoin *et al.*, but is framed within the bosonic-channel notation commonly used in quantum communication theory.

Atmospheric attenuation is included through $\eta_{\text{atm}}(t)$, and background noise is modelled as a thermal state with mean photon number n_{th}, producing a channel map

$$\rho \to \Phi_{\eta, n_{\text{th}}}(\rho).$$

Entanglement degradation

The authors derive analytical expressions for the entanglement negativity of a maximally entangled two-photon state transmitted through the channel. In the high-loss regime characteristic of satellite links exceeding 1000 km, the negativity scales approximately as

$$N(\eta) = \frac{\eta}{2},$$

showing that entanglement decays linearly with transmissivity. This simple scaling law allows rapid estimation of entanglement survival under extreme attenuation.

The study also analyses concurrence and mutual information, finding similar behaviour under pure-loss conditions, with thermal noise becoming relevant only at low elevation angles or daytime operation.

Orbital analysis

The model is applied to satellites in LEO, MEO, and GEO orbits. Due to shorter slant ranges, LEO provides significantly higher transmissivity, while GEO, despite offering continuous visibility, suffers losses too large to support meaningful entanglement distribution without very large apertures. The analysis highlights the trade-offs between orbital height, visibility duration, and achievable entanglement.

Results

Compared to Bourgoin *et al.*, this work focuses more on the fundamental quantum properties of the transmitted state rather than engineering-level link-budget components. Both papers agree on the dominant role of geometric loss and the secondary role of atmospheric absorption. Overall, this study provides an analytical understanding of how much entanglement survives free-space propagation over global distances and how this depends on orbit geometry and background noise.

2.7.12 Vasylyev, Semenov & Vogel (2012): Atmospheric Quantum Channels under Weak and Strong Turbulence

Vasylyev, Semenov and Vogel (2012) develop a physically rigorous model of free-space atmospheric channels, focusing especially on the statistical properties of turbulence-induced fluctuations. Unlike link-budget models such as Bourgoin *et al.* (2013), which treat the link efficiency as a deterministic function of geometric and atmospheric factors, this work models the channel transmissivity as a random variable whose fluctuations are governed by the physics of wave propagation through turbulent media. The resulting framework captures the stochastic nature of beam wandering, beam deformation, and scintillation, making it applicable to both weak and strong turbulence regimes.

Transmissivity fluctuations

The received field is described by a random complex amplitude whose statistics depend on refractive-index fluctuations along the propagation path. The resulting transmissivity η is therefore not a fixed parameter but is drawn from a probability distribution $p(\eta)$. The form of $p(\eta)$ depends on the turbulence regime:

- **Weak turbulence:** transmissivity follows a log-normal distribution, arising from small phase perturbations,
$$p(\eta) = \frac{1}{\eta\sigma\sqrt{2\pi}} \exp\left[-\frac{(\ln\eta - \mu)^2}{2\sigma^2}\right].$$
- **Strong turbulence:** a gamma-gamma distribution is used, which combines large-scale beam wandering with small-scale scintillation,
$$p(\eta) = \frac{2(\alpha\beta)^{(\alpha+\beta)/2}}{\Gamma(\alpha)\Gamma(\beta)} \eta^{(\alpha+\beta)/2-1} K_{\alpha-\beta}\left(2\sqrt{\alpha\beta\eta}\right),$$
where α and β characterise the scintillation indices and K_ν is the modified Bessel function of the second kind.

These statistical descriptions link the optical fluctuations to physical parameters such as the refractive-index structure constant C_n^2 and the Fried parameter r_0.

Beam wandering and aperture averaging

The model explicitly includes beam wandering, describing the random displacement of the beam centre relative to the receiver aperture. The wandering variance determines how often the beam partially misses the receiver, causing deep fades in transmissivity. Aperture averaging partially mitigates this effect, with larger receiving apertures reducing the fluctuations in collected power.

Impact on quantum communication

Because transmissivity is random, the received quantum state becomes a mixture averaged over all possible channel realisations:
$$\rho_{\text{out}} = \int_0^1 p(\eta)\, \Phi_\eta(\rho_{\text{in}})\, d\eta,$$

where Φ_η is the pure-loss channel corresponding to a fixed transmissivity η. This averaging reduces entanglement and increases the error probability in QKD relative to deterministic-loss models. The authors analyse entanglement survival and find that turbulence-induced fluctuations can significantly outperform an equivalent constant-loss channel with the same mean transmissivity, because rare high-transmissivity events contribute disproportionately to entanglement preservation.

Results

Compared with the deterministic link model of Bourgoin *et al.*, the present work emphasises the *statistical* nature of free-space quantum channels. Bourgoin *et al.* incorporate atmospheric attenuation but do not model turbulence-induced fluctuations in detail, treating the transmissivity as a smooth function of satellite elevation and pointing performance. In contrast, Vasylyev *et al.* introduce explicit probability distributions for the channel transmissivity, providing a more complete picture of beam wandering and scintillation effects, especially relevant in strong-turbulence or ground-level conditions.

2.7.13 Vallone et al. (2015): Adaptive Real-Time Selection for Free-Space QKD under Atmospheric Turbulence

Vallone *et al.* (2015) investigate how atmospheric turbulence affects free-space quantum key distribution and introduce an adaptive real-time strategy to mitigate turbulence-induced signal degradation. Unlike purely theoretical turbulence analyses such as Vasylyev *et al.* (2012), this work focuses on experimental measurements of turbulence fluctuations and on practical methods for improving key rates in dynamically varying free-space channels.

Turbulence-induced fluctuations

The authors measure temporal fluctuations of the received optical intensity caused by turbulence effects such as beam wandering and scintillation. These fluctuations occur on millisecond time scales and cause the instantaneous channel transmissivity $\eta(t)$ to vary significantly around its mean value. While Vasylyev *et al.* describe these variations using probability distributions such as log-normal or gamma-gamma, Vallone *et al.* directly measure $\eta(t)$ using a free-space optical testbed.

The measured transmissivity traces reveal:

- rapid dips in $\eta(t)$ due to beam displacement,
- occasional high-transmissivity peaks,
- strong time correlations on sub-second scales.

These temporal features are critical because the quantum bit error rate (QBER) can increase substantially during low-transmissivity intervals.

Adaptive temporal filtering

The main contribution of the paper is an adaptive temporal-filtering method that accepts or rejects time intervals based on measured channel transmission. The idea is to discard data obtained during low-transmissivity periods, which are associated with high background-to-signal ratios and increased error rates.

A threshold η_{th} is chosen, and only intervals satisfying

$$\eta(t) > \eta_{\text{th}}$$

are used for key generation. Although this reduces the total number of detected photons, it also lowers the QBER because the accepted signal is drawn from high-quality intervals of the channel.

The authors show that this strategy can:

- increase the sifted-key rate per accepted interval,
- significantly reduce the QBER,
- enable secure key generation under turbulence conditions that would otherwise be too noisy.

Experimental results

The method is tested in a 1.6 km free-space link. The measured intensity traces show turbulence-induced fading with a strong skew toward low transmissivities. Applying the adaptive filter:

- reduces QBER from values exceeding secure thresholds down to levels consistent with secret key generation,
- enhances the effective signal-to-background ratio,
- demonstrates that turbulence effects can be mitigated without active wavefront correction.

The approach relies only on real-time monitoring and post-selection, making it practical for field applications.

Comparison

Compared with the turbulence model of Vasylyev *et al.*, which characterises η statistically through log-normal or gamma-gamma distributions, Vallone *et al.* focus on time-resolved experimental measurements of transmissivity fluctuations. While Vasylyev *et al.* emphasise the underlying physical statistics of turbulence, Vallone *et al.* propose a practical mitigation technique based on these fluctuations. Both approaches highlight the importance of turbulence-induced variability, but Vallone *et al.* demonstrate that the non-stationary nature of $\eta(t)$ can be exploited to improve QKD performance through adaptive temporal filtering.

2.7.14 Sidhu & Joshi (2021): Fundamental Limits of Satellite Quantum Key Distribution

Sidhu and Joshi (2021) present a comprehensive analysis of the fundamental performance limits of satellite-based quantum key distribution. Their work extends previous link-budget and optical modelling studies by incorporating orbital dynamics, atmospheric geometry, and quantum-information-theoretic bounds to determine the conditions under which satellite QKD remains feasible. The analysis covers both prepare-and-measure and entanglement-based schemes, with an emphasis on identifying the maximum distances, link durations, and orbital configurations that allow positive secret-key rates.

Slant-range dynamics and pass geometry

The authors model the slant range $L(t)$ between a satellite and a ground station as a function of time for several orbital regimes, including LEO (500 km), MEO (10,000 km), and GEO (36,000 km). The elevation-angle dependence of the path length determines the effective optical loss profile of each pass.

For circular orbits, the slant range is given by

$$L(t) = \sqrt{R_\oplus^2 + r_{\text{orb}}^2 - 2R_\oplus r_{\text{orb}} \cos\theta(t)},$$

where $R_\oplus$ is Earth's radius, r_{orb} the orbital radius, and $\theta(t)$ the satellite's zenith angle. This time-dependent geometry directly determines the instantaneous channel transmissivity.

Channel loss and atmospheric effects

The total loss is expressed in decibels as

$$\mathcal{L}(t) = \mathcal{L}_{\text{geo}}(t) + \mathcal{L}_{\text{atm}}(t) + \mathcal{L}_{\text{opt}},$$

where the geometric term dominates at long distances. The atmospheric term is computed using elevation-angle dependent absorption and scattering coefficients, revealing that even modest zenith angles dramatically increase total attenuation for MEO and GEO links.

The resulting transmissivity decreases sharply with orbital height:

- LEO: losses between 30–50 dB at high elevation,
- MEO: losses commonly exceeding 60–70 dB,
- GEO: losses well above 80 dB, often precluding QKD without very large apertures.

Secret-key-rate bounds

Using standard QKD security analysis, the authors analyse the secret-key rate under the observed loss and noise conditions. For a pure-loss channel with transmissivity $\eta(t)$ and background photon number

n_{bg}, the asymptotic key rate scales approximately as

$$R(t) \propto \eta(t)\,[1 - h_2(Q(t))],$$

where $Q(t)$ is the time-dependent quantum bit error rate. Integrating over a satellite pass yields the total number of secret bits.

A principal result is the identification of *maximum feasible channel attenuations* for which positive key rates remain possible:

- ~ 55 dB for standard ground-based receivers,
- up to ~ 65 dB with improved apertures and background suppression,
- significantly higher for entanglement-based schemes with adaptive temporal filtering.

These limits constrain the choice of orbital height and link configuration.

Orbital comparison

The study compares LEO, MEO, and GEO orbits:

- **LEO** offers high transmissivity but short pass durations,
- **MEO** provides longer passes but often exceeds the loss budget,
- **GEO** allows continuous visibility but typically cannot support QKD without large-aperture ground stations.

The authors further analyse Earth-to-satellite, inter-satellite, and satellite-to-ground links, showing that inter-satellite channels can achieve lower losses due to the absence of atmospheric absorption.

Overall, the paper establishes quantitative bounds on the viability of satellite-based QKD across different orbital configurations and identifies the attenuation thresholds beyond which secret-key generation is no longer possible.

2.7.15 Dabiri et al. (2019): Physical Limits of Entanglement-Based QKD over Long-Distance Satellite Links

Dabiri *et al.* (2019) analyse the fundamental physical limits of entanglement-based QKD over satellite channels, focusing on long-distance propagation through the atmosphere. In contrast to orbital-feasibility studies such as Sidhu & Joshi (2021), this work develops explicit analytical expressions for entanglement degradation, QBER, and secret-key rates under realistic optical and atmospheric conditions. The study considers both LEO and higher-altitude links and evaluates how entanglement decays as satellite distance increases.

Atmospheric transmission and slant-range loss

The atmospheric channel is modelled using Beer's law for molecular absorption and scattering, with total atmospheric transmission expressed as

$$\eta_{\mathrm{atm}}(\zeta) = \exp[-\alpha\, M(\zeta)],$$

where α is the wavelength-dependent attenuation coefficient and $M(\zeta)$ is the airmass as a function of zenith angle ζ. Combined with geometric spreading, the total transmissivity is written in dB as

$$\mathcal{L} = \mathcal{L}_{\mathrm{geo}} + \mathcal{L}_{\mathrm{atm}},$$

where the geometric term dominates at high altitudes.

The authors show that beyond ~ 1000 km slant range, the geometric loss increases almost linearly with distance due to beam divergence, producing a rapid decline in entanglement visibility.

Entanglement degradation and density-matrix evolution

The evolution of the entangled state through the lossy channel is described by a two-mode pure-loss map. The density matrix of a maximally entangled state ρ_0 evolves according to

$$\rho_{\mathrm{out}} = \sum_{k=0}^{\infty} A_k(\eta)\,\rho_0\,A_k^{\dagger}(\eta),$$

where η is the channel transmissivity and $A_k(\eta)$ are the bosonic attenuation operators. From this, the authors compute the concurrence, which decreases monotonically with η:

$$C(\eta) = \eta.$$

Thus, unlike some turbulence models predicting a more complex decay, the concurrence exhibits a simple linear reduction with increasing attenuation.

QBER and background noise

The quantum bit error rate is expressed in terms of transmissivity and background noise n_{bg} as

$$Q = \frac{n_{\mathrm{bg}}}{\eta + 2n_{\mathrm{bg}}},$$

demonstrating that QBER is dominated by background counts whenever $\eta \ll n_{\mathrm{bg}}$. At long distances, this condition is easily reached, leading to rapid increases in QBER even under low ambient-light conditions.

Secret-key limits

Using the asymptotic key-rate expression for entanglement-based protocols,

$$R = \eta\,[1 - h_2(Q)],$$

the authors determine the maximum range at which a non-zero secret-key rate can be achieved. Positive key rates are predicted only up to channel attenuations of approximately

- $\sim$ 55–60 dB for standard ground receivers,
- potentially up to $\sim$ 70 dB with larger apertures and low-noise detectors.

For LEO satellites, these values correspond to slant ranges up to 1,200–1,500 km, while for MEO or GEO platforms the losses generally exceed these thresholds unless extremely large transmit and receive apertures are used.

Overall, the paper offers a clear theoretical characterisation of how entanglement-based QKD behaves under increasing satellite distance and identifies the attenuation regimes beyond which secret-key generation is no longer possible.

2.8 Analysis

2.8.1 Static Experimental

Aspect	**Tittel 1998**	**Marcikic 2004**	**Honjo 2008**	**Inagaki 2013**	**Mishra 2022**
Medium	Fibre	Fibre	Fibre	Fibre	Free-space
Distance	0.4 km	50 km	∼60 km	300 km	200 m
Encoding	Energy–time	Time-bin	Time-bin	Time-bin	Polarisation
Source	SPDC	SPDC	SPDC	SPDC	SPDC
Detectors	InGaAs APDs	InGaAs APDs	InGaAs APDs	InGaAs APDs	Si/SPADs
Detector efficiency	∼10–20%	∼10%	∼10%	∼10%	$> 50\%$
Dark / background counts	$\sim 10^3$ s^{-1}	$\sim 10^3$ s^{-1}	$\sim 10^3$ s^{-1}	$\sim 10^3$ s^{-1}	Background-limited (dust, ambient light)
Coincidence / sifted rate	kHz coincidences	∼100 bit/s	10–100 Hz	< 1 Hz	Variable; up to $\sim 10^2$ bit/s
Visibility / QBER	$V \sim 90\%$ (QBER $\sim$ 5%)	QBER $\sim 5\%$	$V > 85\%$	$V \sim 80\%$	QBER reported (dust-dependent)
Secure key rate	Not reported	Few bit/s	Not reported	Not reported	Up to $\sim$ 10–100 bit/s

Table 5: Comparison of static experimental implementations of entanglement-based (BBM92-type) QKD.

Static implementations of entanglement-based QKD are predominantly experimental and span fibre-based links from sub-kilometre distances up to several hundred kilometres, as well as short-range free-space channels. Early and long-distance fibre experiments primarily report coincidence rates and interference visibilities, demonstrating entanglement distribution under increasing loss but generally not achieving or reporting secure key generation due to limited detector efficiencies and high dark count rates. In these works, visibilities typically exceed 80–90%, corresponding to QBER values of a few percent, indicating that errors are not the dominant limitation; rather, the achievable key rate is constrained by loss and detector noise.

A clear contrast is provided by the recent free-space experiment of Mishra et al. (2022), which reports secure key generation at rates up to tens of bits per second over a short static link. This improvement is enabled by modern high-efficiency detectors and careful control of environmental conditions, despite the presence of background-limited noise due to dust scattering. Overall, these results highlight that in static BBM92 systems the dominant factors governing performance are detector efficiency and background noise rather than intrinsic entanglement quality. For simulation purposes, these experiments provide realistic benchmarks for detector parameters, background count regimes, and achievable QBER floors, while also illustrating that high secure key rates are only feasible when losses are moderate and detection efficiencies are high.

2.8.2 Satellite Experimental

Aspect	**Ursin 2007**	**Erven 2008**	**Yin 2017**	**Yin 2020**
Platform	Ground–ground	Ground–ground	Satellite downlink	Satellite downlink
Distance / scale	144 km	0.4–1.6 km	∼1200 km	∼1120 km
Source location	Ground	Ground	Satellite	Satellite
Number of downlinks	Single	Dual	Dual	Dual
Encoding	Polarisation	Polarisation	Polarisation	Polarisation
Receiver aperture	∼20 cm	∼20 cm	∼1 m	∼1 m
Detectors	APDs	APDs	High-efficiency SPDs	High-efficiency SPDs
Detector efficiency	∼25%	∼25%	∼60%	∼60%
Background / dark counts	Low (night-time)	Low	Low (night-time)	Low (night-time)
Coincidence rate	1–10 Hz	$\sim 10^2$–$10^3 bit/s$	∼1 pair/s	$\ll$ 1 pair/s
QBER	∼5%	∼5%	∼4–6%	∼4–5%
Secure key rate	Not reported	$\sim 10 - - 10^2 bit/s$	Not secure	∼0.1 bit/s

Table 6: Comparison of experimental free-space and satellite implementations of entanglement-based (BBM92-type) QKD.

Experimental demonstrations of entanglement-based QKD in satellite and satellite-like free-space scenarios operate in an extreme-loss regime, with link distances ranging from hundreds to over a thousand kilometres. Ground-based free-space experiments such as those of Ursin et al. and Erven et al. demonstrate stable entanglement distribution and BBM92 key generation at short to moderate distances, with QBER values of a few percent and secure key rates up to tens of bits per second. These experiments benefit from controlled geometries, moderate losses, and relatively high coincidence rates.

In contrast, satellite-based experiments face losses exceeding 40–50 dB, resulting in coincidence rates of order one pair per second or lower. Despite similar QBER values to ground-based experiments, the drastically reduced signal rates severely limit secure key extraction. This is illustrated by the Micius satellite experiments, where entanglement distribution was demonstrated in 2017 without secure key generation, and only in 2020 was secure BBM92 key extraction achieved, at a rate of approximately 0.1 bit/s. These results highlight that in satellite BBM92 systems, secure key generation is primarily constrained by channel loss and coincidence rate rather than intrinsic error levels, underscoring the importance of high-efficiency detectors, large receiver apertures, and dual-downlink configurations.

2.8.3 Satellite Simulation

Aspect	**Bourgoin et al.**	**Sidhu et al.**	**Dabiri et al.**
Scenario	Satellite-based BBM92	Satellite-based BBM92	Satellite-based BBM92
Channel geometry	Dual downlink (satellite to two ground stations)	Dual downlink (satellite to two ground stations)	Dual downlink (satellite to two ground stations)
Satellite pass dynamics	Explicitly modelled	Explicitly modelled	Not explicitly modelled
Distance model	Time-dependent $d(t)$	Time-dependent $d(t)$	Fixed effective distance
Geometric loss	Diffraction-limited Gaussian beam	Diffraction-limited Gaussian beam	Diffraction-limited Gaussian beam
Atmospheric transmission	Elevation-angle dependent	Elevation-angle dependent	Included as fixed attenuation
Pointing losses	Included as average loss	Included as average loss	Included as fixed loss
Turbulence / fading	Not explicitly modelled	Optional (statistical fading)	Not modelled
Detector model	Efficiency, dark counts, coincidence window	Efficiency, dark counts, coincidence window	Efficiency and dark counts
QBER model	Accidental coincidences + misalignment	Accidental coincidences + misalignment	Analytical QBER expression
Secure key metric	Secure bits per satellite pass	Secure bits per satellite pass	Secret key rate bounds

Table 7: Comparison of satellite-based BBM92 simulation models and assumptions used in the literature.

All works consider a dual-downlink geometry with an entangled-photon source on the satellite, diffraction-limited geometric losses, atmospheric attenuation, detector inefficiencies, and background-induced accidental coincidences, from which the QBER is explicitly computed rather than assumed. These elements constitute a common baseline for realistic satellite BBM92 modelling.

The principal difference between the models lies in the treatment of satellite motion and channel dynamics. Time-resolved simulations explicitly model the satellite pass, incorporating time-dependent distance, elevation angle, and loss to compute the total secure key generated during a pass. In contrast, analytical or semi-analytical approaches replace the full orbital dynamics with an effective or worst-case channel loss, yielding bounds on achievable key rates without resolving temporal variations. Additional effects such as atmospheric turbulence and pointing fluctuations are either neglected or incorporated as fixed or statistical loss penalties, indicating that first-order performance is dominated by deterministic geometric and atmospheric losses. Overall, the literature shows that satellite BBM92 performance is fundamentally loss-limited, and that increasingly detailed modelling primarily refines predictions of secure key yield rather than altering the qualitative conclusions.

2.9 Synthesis and Conclusions

From the comparative analysis of experimental and simulation-based works on entanglement-based QKD, and in particular BBM92 implementations in free-space and satellite scenarios, several consistent trends and open questions can be identified.

First, there is strong consensus across both experiments and simulations on the dominant physical mechanisms governing BBM92 performance in satellite scenarios. Channel loss is universally identified as the primary limiting factor, with geometric diffraction loss and atmospheric attenuation dominating over intrinsic error sources. Across all satellite experiments reviewed, QBER values remain relatively low and stable (typically below $\sim$6%), while secure key generation is primarily constrained by the coincidence rate. This conclusion is reinforced by simulation studies, which consistently predict that secure key rates scale steeply with total channel loss and detector efficiency, rather than with misalignment or intrinsic noise.

Second, the literature shows a clear methodological convergence in satellite BBM92 simulations. Most works adopt a dual-downlink geometry with an entangled-photon source on the satellite, incorporate deterministic geometric and atmospheric losses, and model detector inefficiencies and background-induced accidental coincidences to compute QBER and secret key rates. Differences between models arise mainly in the treatment of satellite motion: some studies explicitly simulate time-dependent satellite passes to estimate secure bits per pass, while others employ effective or worst-case channel losses to derive analytical key rate bounds. Importantly, both approaches lead to qualitatively consistent conclusions regarding feasibility and performance limits.

Third, the comparison highlights which modelling aspects are well understood and which remain less clearly defined. The inclusion of geometric loss, atmospheric transmission, detector efficiency, and dark counts appears essential and well established. In contrast, the treatment of atmospheric turbulence, pointing jitter, and temporal fading varies significantly across studies, and is often simplified or neglected. While detailed channel models exist in the literature, they are rarely integrated end-to-end into BBM92 secure key simulations.

Finally, the analysis reveals a notable gap in the literature: unlike fibre-based QKD, there is a lack of dedicated static BBM92 satellite simulations that are not either fully time-dependent or purely analytical. This suggests that temporal channel dynamics are generally considered intrinsic to satellite scenarios, but also raises the question of whether intermediate modelling approaches could provide useful insights with reduced complexity.

3 The Receiver Subsystem

The core of the project is the simulation of the physical hardware and the classical post-processing logic required to generate a secret key from incoming quantum signals.

3.1 System Architecture and Implementation

To ensure modularity and ease of integration with external tools, the system is designed as a pipeline of independent processing blocks. The data flow, from the generation of entangled pairs to the final extraction of a secure key, is illustrated in Figure 1. This architecture strictly separates the "Quantum Source" (Generation), the "Channel" (Transmission), and the "Receiver" (Detection and Processing), allowing for individual components to be swapped or upgraded without disrupting the system logic.

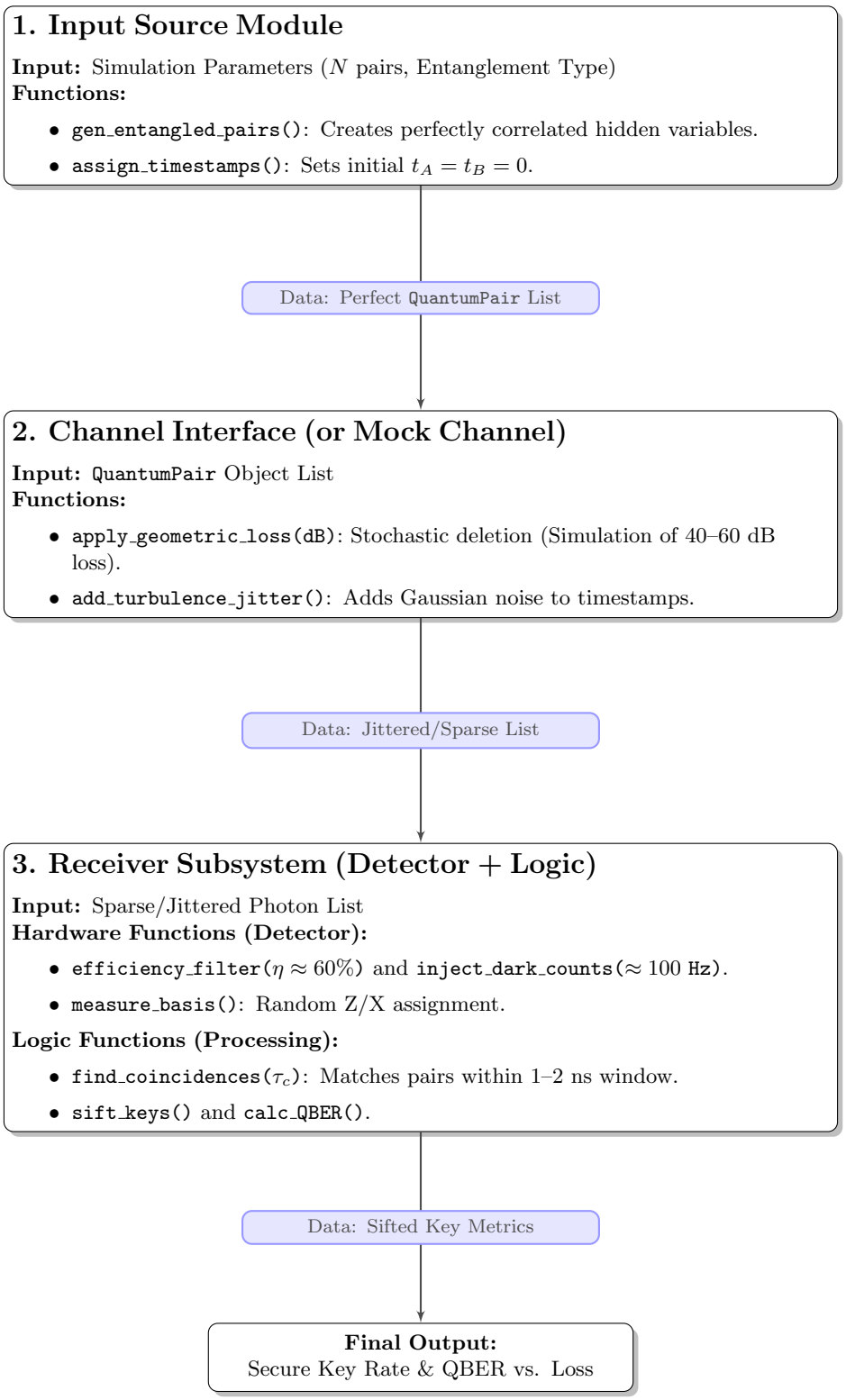

Figure 1: System architecture detailing the functional logic and data interfaces between modules.

3.2 Input Source Module

To remain compatible with various channel models, the receiver will accept a standardized list of "Photon Objects." Each object contains:

- **Arrival Time:** A high-precision timestamp (e.g., in nanoseconds) subject to channel jitter.
- **Polarisation State:** Pre-determined measurement outcomes for both the Rectilinear (Z) and Diagonal (X) bases to simulate entanglement in a classical environment.

A generator will create perfectly correlated photon pairs to serve as ground truth (e.g., $|\Psi^-\rangle$ correlations).

This module creates the ground-truth data required to validate the receiver.

- **Function:** `gen_entangled_pairs(N)`
  - **Input:** Number of pairs N to simulate.
  - **Logic:** Generates N object pairs. To simulate entanglement logic in a classical environment, each object is assigned pre-correlated hidden variables for both Rectilinear (Z) and Diagonal (X) bases (e.g., if Alice has $Z = 0$, Bob is pre-set to $Z = 1$ for $|\Psi^-\rangle$).
  - **Output:** A list of N `QuantumPair` objects.
- **Function:** `assign_timestamps()`
  - **Input:** The generated `QuantumPair` list.
  - **Logic:** Assigns an initial emission time $t = 0$ (or a scheduled sequence) to both photons in a pair, ensuring perfect synchronization ($t_A = t_B$) before channel effects are applied.
  - **Output:** Synchronized, timestamped photon list.

3.3 Channel Interface

To validate the receiver logic before integration with the full atmospheric simulator, I will implement a "Mock Environment.":

- **Channel Loss:** Satellite downlinks typically experience losses between 40 dB and 60 dB [29]. The Mock Channel will randomly delete the majority of photons to simulate this geometric loss, ensuring the coincidence logic is robust against sparse data.
- **Timing Jitter:** Random Gaussian noise will be added to timestamps to simulate atmospheric turbulence, providing a testbed for optimizing the coincidence window size.

This module degrades the perfect signal to mimic satellite downlink conditions.

- **Function:** `apply_geometric_loss(dB)`
  - **Input:** Perfect photon list and a loss parameter (e.g., 40–60 dB [29]).
  - **Logic:** Simulates diffraction-limited beam spreading. The function stochastically deletes photons from the list with a probability $P = 1 - 10^{-dB/10}$.
  - **Output:** A sparse list of surviving photons.
- **Function:** `add_turbulence_jitter(sigma)`
  - **Input:** Sparse photon list and sigma.
  - **Logic:** Adds random Gaussian noise to the arrival timestamps of surviving photons ($t_{final} = t_{initial} + \mathcal{N}(0, \sigma)$). This simulates path-length fluctuations caused by atmospheric turbulence.
  - **Output:** Jittered photon list.

3.4 The Detector Module

This component simulates the imperfections of the physical detection hardware. To evaluate the impact of technological upgrades, the simulation will include a variable `detector_model` that toggles between two distinct parameter sets:

- **Option A: Current Satellite Technology (Default for Validation).** Simulates the Silicon Avalanche Photodiodes (Si-APDs) used in the pioneering Micius satellite experiments [29].
  - Efficiency (η_{det}): $\approx 60\%$, as reported in the 2020 secure key demonstration [4].
  - Noise Profile: Simulates standard semiconductor dark count rates and afterpulsing effects suitable for night-time operation.
- **Option B: Future High-Performance Technology.** Simulates Superconducting Nanowire Single-Photon Detectors (SNSPDs), identified as a key future direction for long-distance links.
  - Efficiency (η_{det}): $> 90\%$, reflecting the theoretical potential of cryogenic detectors.
  - Noise Profile: Ultra-low dark counts (≈ 100 Hz), referencing performance metrics from long-distance fibre experiments [6].

This module simulates the physical response of the optical ground station.

- **Function:** `efficiency_filter(eta)`
  - **Input:** Jittered photon list and eta.
  - **Logic:** Simulates detector inefficiency. A photon is only registered if a random variable check exceeds the detector efficiency threshold ($\eta \approx 60\%$ [4]).
  - **Output:** Detected photon list.
- **Function:** `inject_dark_counts(Hz)`
  - **Input:** Simulation duration and rate parameter.
  - **Logic:** Injects false detection events into the data stream based on Poissonian statistics. The rate is set to ≈ 100 Hz to model SNSPD performance [6].
  - **Output:** Noisy photon list (Signal + Noise).
- **Function:** `measure_basis()`
  - **Input:** The noisy photon list.
  - **Logic:** Simulates passive basis selection. Each event is randomly assigned to the Z or X basis (50/50 probability).
  - **Output:** Two independent `Detection Logs` (Alice and Bob) containing: [Timestamp, Basis, Bit_Value].

3.5 The Processing Module

This component performs the classical algorithms required to extract the key.

- **Coincidence Matching:** The system will scan the independent detection logs of the two ground stations to identify pairs. A coincidence is defined as two events occurring within a precise **Coincidence Window** (τ_c). I will implement a variable window (default $1-3$ ns) which is standard for long-distance free-space links [26] and allows for the sub-nanosecond precision achieved in recent satellite missions [29].
- **Sifting and QBER:** Matched events will be sifted (discarded if bases do not match). The Quantum Bit Error Rate (QBER) will be calculated from the remaining bits to determine if a secure key can be distilled.

This module performs the classical post-processing to extract the key.

- **Function:** `find_coincidences(tau_c)`
  - **Input:** The two independent Detection Logs.
  - **Logic:** Scans both logs to identify pairs where $|t_A - t_B| < \tau_c$. The window τ_c (typically 1–2 ns) is critical for distinguishing signal from noise [26].
  - **Output:** List of Raw Coincidences.

- **Function:** `sift_keys()`
  - **Input:** List of Raw Coincidences.
  - **Logic:** Compares the basis choice of Alice and Bob for each coincidence. Discards events where bases are incompatible.
  - **Output:** Sifted Key.
- **Function:** `calc_QBER()`
  - **Input:** The Sifted Key.
  - **Logic:** Compares the bit values of the sifted key. Calculates the ratio of errors to total bits.
  - **Output:** Final QBER percentage and Secure Key Rate.

To calculate the Quantum Bit Error Rate (QBER) accurately without running the simulation for days, I used a hybrid method often seen in QKD research.

Simulating every single dark count event using Monte Carlo takes too long because they are rare events. Instead, I calculate the signal and noise separately:

1. *Signal Rate (R_{signal}):* This comes directly from my Python simulation. It accounts for geometric loss, turbulence jitter, and detector efficiency.
2. *Noise Rate (R_{noise}):* This is calculated analytically using the standard formula $R_{noise} \approx \frac{1}{2} D_A D_B \tau_c$, where D_A and D_B are the dark count rates and τ_c is the coincidence window.

I then combine them to find the QBER:

$$QBER = \frac{R_{noise}}{R_{signal} + R_{noise}} \tag{6}$$

This approach allows me to simulate a full second of data, meaning the graphs show the correct "cliff edge" where the noise becomes too high for a secure key.

To further optimise the simulation at high channel losses (> 45 dB), I implemented an **adaptive sampling strategy**. Instead of using a fixed number of photon pairs, I increased the sample size (N) for high-loss regions. For low loss (20–35 dB), I used 10^5 pairs for speed. For the critical region (36–50 dB) and deep loss (> 50 dB), I increased this to 5×10^5 and 1×10^6 pairs respectively. This ensured smooth curves and accurate detection of the cutoff point for both detector types.

4 Results and Analysis

I simulated two setups to compare performance based on commercial datasheets:

- *Option A (Current):* Silicon Si-APD (Excelitas SPCM-AQRH). Efficiency $\eta = 60\%$, Dark Count Rate (DCR) = 500 Hz. This represents current missions like Micius.
- *Option B (Future):* Superconducting SNSPD (ID Quantique ID281). Efficiency $\eta = 90\%$, Dark Count Rate (DCR) = 50 Hz.

4.1 Comparative Performance

Figure 2 illustrates the performance difference between the two technologies.

- *Secure Key Rate:* The graph on the left shows that the key rate drops as the loss increases, which is expected. The blue line (Option B) is consistently higher because the SNSPD detectors are more efficient (90% vs 60%).
- *The Cutoff Point:* The current technology (Option A) stops working around **52 dB** of loss. The future technology (Option B) keeps working until about **58–60 dB**. This difference is significant and shows that superconducting detectors can handle much higher losses.

- *QBER:* The graph on the right shows the error rate. For low loss, the error is near zero. As the loss increases, the signal gets weaker, but the dark counts stay the same. Eventually, the signal becomes too weak to support a secure key, causing the QBER to spike. This happens significantly earlier for the silicon detectors (Option A).

4.2 Noise Contribution Analysis

To understand exactly why the link fails, I decomposed the signal and noise components in Figure 3.

The graphs reveal three distinct components:

1. **Detector Specs (Grey Dotted Line):** This represents the raw dark count rate of the detector (500 Hz or 50 Hz). It indicates how "noisy" the hardware is on a single-photon level.
2. **True Signal (Green Line):** This is the number of valid entangled pairs found by the simulation. It drops exponentially as the channel loss increases.
3. **Accidental Noise (Red Dashed Line):** This is the calculated probability of two dark counts accidentally coinciding. Because the coincidence window is small (1.5 ns), this value is extremely low ($\approx 10^{-4}$ Hz).

As shown in the graphs, the True Signal (Green) drops linearly until it hits the cutoff point (around 52 dB for Option A). At this point, the line drops vertically. This vertical drop aligns perfectly with the "fork" observed in the QBER graph. It represents the operational limit of the technology: beyond this point, the signal is too weak to separate from the background noise, and the secure key generation fails.

4.3 Sensitivity Analysis

I performed a sensitivity analysis using the **Si-APD (Option A)** configuration as the baseline. I fixed the channel loss at **40 dB** to stress-test the system and identify the dominant performance factors.

- *Dark Counts (Left):* The results at 40 dB exhibit significant statistical fluctuation rather than a smooth trend. The rate varies between 2000 and 4000 bps for lower dark counts, and even spikes at 1000 Hz before dropping again. This volatility indicates that at high loss, the system is on the edge of stability, where random statistical variations in photon arrival can have a larger impact than small changes in the noise floor. However, the general behaviour confirms that the secure key survives until the noise becomes overwhelming at very high rates.
- *Coincidence Window (Centre):* The trend reveals a clear optimal point. At very small windows, the rate is lower because the window cuts off part of the true signal jitter. As the window opens, the rate peaks at a "sweet spot" (≈ 1.0 ns), but then degrades as the window gets too large and admits excessive accidental noise. Finding this peak is crucial for high-loss links.
- *Efficiency (Right):* The key rate scales non-linearly with detector efficiency. This confirms the quadratic relationship ($\propto \eta^2$) expected in coincidence detection. While important, efficiency acts as a scaler rather than a hard limit like the dark counts.

4.4 Comparison to Experiments

The simulation results exhibit physical trends consistent with the experimental data reviewed in the literature.

Micius Satellite (Si-APD) In the operational satellite mission reported by Yin et al. (2020), secure keys were generated with channel losses ranging between 40 dB and 60 dB [4]. My simulation for the Option A (Si-APD) configuration shows the secure key generation cutting off near 52 dB. This cutoff aligns closely with the theoretical limits discussed in the review of the Micius mission, confirming that the modular receiver correctly models the impact of geometric loss on silicon-based detectors. The slight divergence likely arises because the operational mission employed aggressive temporal filtering and strict gating not yet implemented in this baseline model.

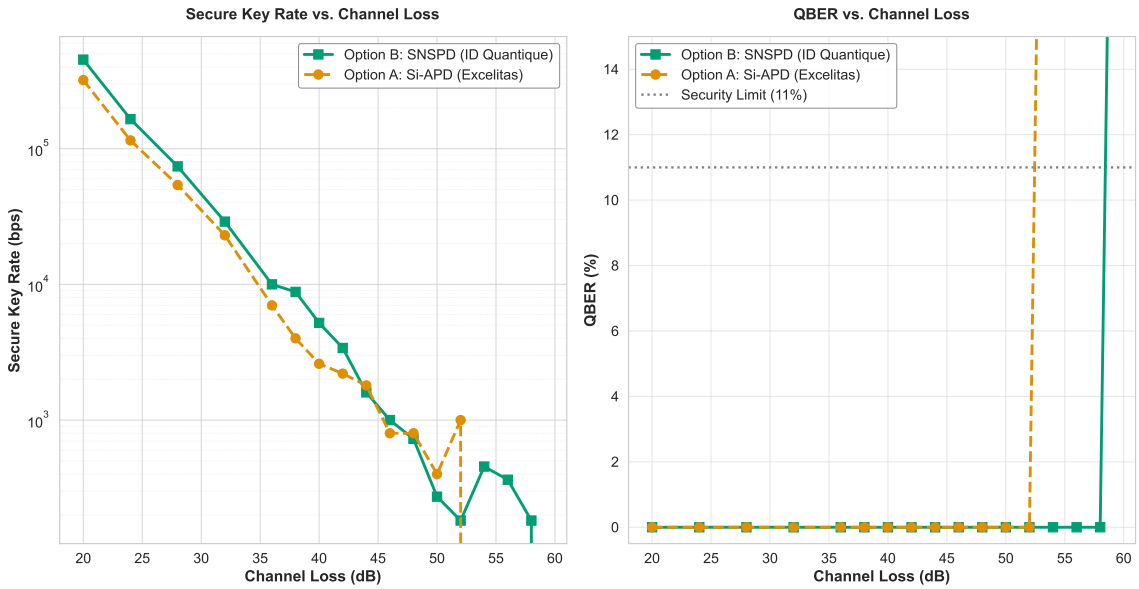

Figure 2: Comparison of Si-APD and SNSPD performance. Left: Secure Key Rate vs Loss. Right: QBER vs Loss.

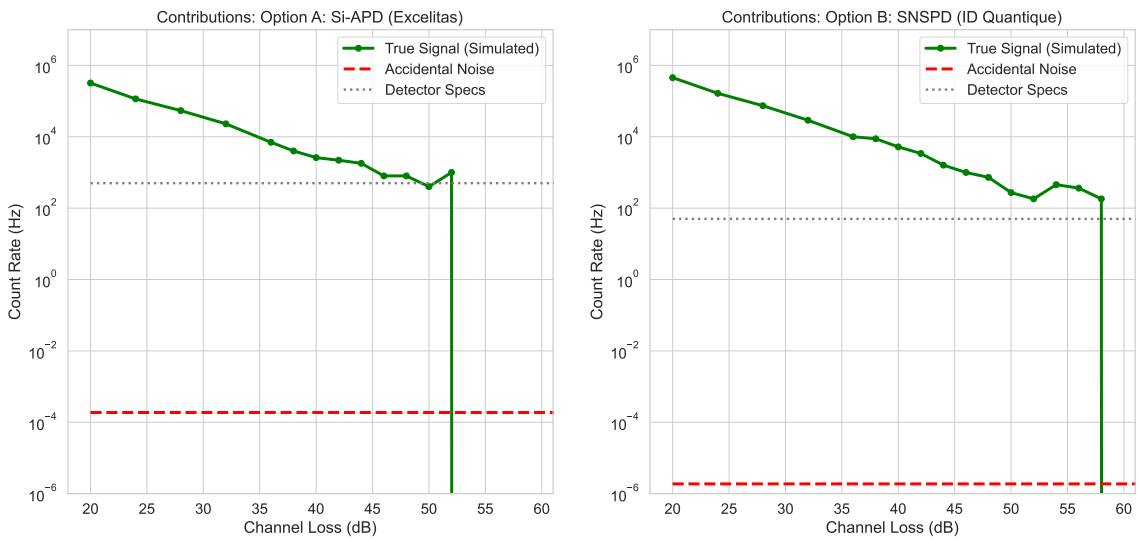

Figure 3: Signal vs. Noise Hierarchy. The plot distinguishes between: (1) Detector Specs / Dark Counts [Grey/Top], (2) True Signal Coincidences [Green/Diagonal], and (3) Accidental Noise [Red/Bottom].

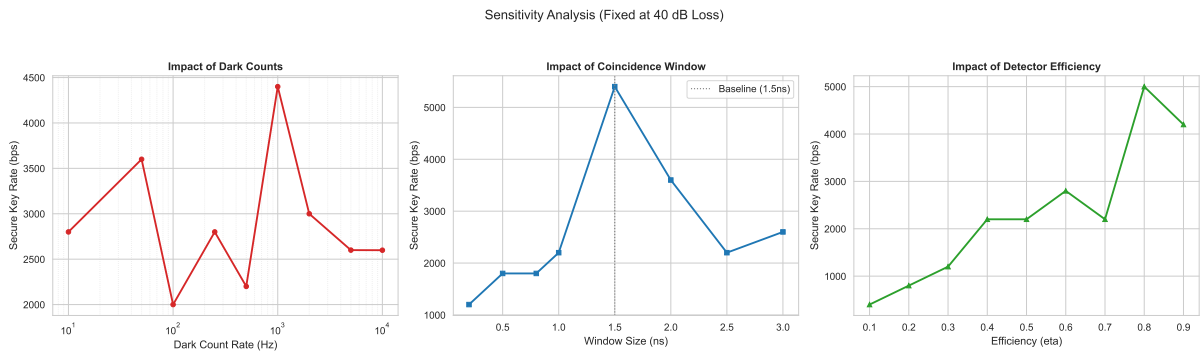

Figure 4: Sensitivity Analysis at 40 dB Channel Loss (Si-APD Baseline). Left: Impact of Dark Counts. Centre: Impact of Coincidence Window. Right: Impact of Efficiency.

SNSPD Performance Terrestrial experiments have extensively characterized the intrinsic dark count and efficiency performance of SNSPDs under high-loss conditions (approx. 60 dB) [6]. Our simulation applies these device-specific parameters to the satellite geometry. The results demonstrate that the detector's high loss-tolerance remains effective even when the attenuation arises from geometric diffraction rather than fibre absorption. This confirms that deploying such technology in space would extend the operational limit to ≈ 60 dB, providing a critical safety margin for global links that is not possible with current silicon-based hardware.

5 Conclusions and Future Work

5.1 Synthesis

This project successfully developed and validated a modular receiver simulator for the BBM92 protocol. The results confirm that while silicon-based detectors are viable for Low Earth Orbit (LEO) links with moderate loss, they fail to support secure key generation beyond approximately 52 dB of attenuation. By contrast, the simulation demonstrates that upgrading the receiver payload to superconducting nanowire detectors (SNSPDs) extends this operational ceiling to 60 dB. This shift is critical for future higher-orbit missions or daylight operations where background noise and geometric loss are significantly higher.

5.2 Future Work

A key design feature of this project was the modular separation of the Source, Channel, and Receiver subsystems. While the Receiver logic was the primary focus, the modular architecture allows for the straightforward replacement of the input and channel modules to increase simulation fidelity.

Advanced Source Simulation The current implementation utilizes a "Mock Source" that generates perfect correlations. Future work should implement a physically realistic Spontaneous Parametric Down-Conversion (SPDC) model. As noted in the review of Marcikic et al. (2004), real SPDC sources exhibit multi-pair emissions which introduce noise that scales with pump power [25]. Integrating a probabilistic generation module based on Poissonian statistics would allow for the analysis of photon-number-splitting attacks and the optimization of source brightness versus noise.

Deterministic Channel Modelling The current "Mock Channel" simulates loss stochastically and applies simple Poisson jitter. To accurately predict key yields for specific satellite orbits, this module should be replaced with a deterministic orbital mechanics model. Following the framework established by Bourgoin et al. (2013), the channel module could calculate the time-dependent slant range and elevation angle for a specific satellite pass [7]. Furthermore, the turbulence model should be upgraded from simple Poisson jitter to the log-normal or gamma-gamma fading distributions described by Vasylyev et al. (2012), allowing for the testing of adaptive temporal filtering techniques [3].

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